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Tumor-resident T cells and dendritic cells form an in situ archetype during immunotherapy response in melanoma

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Tumor-resident (TR) T cells, known as tissue-resident memory (TRM) T cells in mice, play a central role in melanoma immunosurveillance, yet their contribution to immune checkpoint inhibitor (ICI) therapy has not been comprehensively explored. We performed spatial and single-cell profiling on 32 metastatic melanoma lymph node samples, from treatment-naïve, ICI-resistant and ICI-responsive patients. Here we show that tumor areas in ICI-responders were enriched for both $CD8^+$ and $CD4^+$ TR. $CD8^+$ TR cells were clonally expanded, and both $CD8^+$ and $CD4^+$ TR cells upregulated cytotoxicity-related gene expression, suggesting functional anti-tumor immunity. Conversely, ICI-resistant tumors displayed chronic IFN-γ response pathways, linked to T cell exhaustion. We further identified a spatially organized immune triad composed of $CD8^+$ TR, $CD4^+$ TR, and type-3 dendritic cells (DC3) that is exclusive to responding tumors. These findings define coordinated cellular interactions within the tumor microenvironment that underpin successful immunotherapy and provide a framework for spatial biomarkers of response.

Melanoma is a highly immunogenic tumor type, which strongly influences response to immune checkpoint inhibitor (ICI) therapies[1–3]. Large-scale sequencing studies show that tumors rich in tumor-infiltrating lymphocytes (TILs) are associated with a favorable prognosis[4–8]. In murine models, tissue-resident memory (TRM) T cells play a central role in melanoma immune surveillance and control of tumor progression[9,10]. TRM-mediated melanoma killing in the skin induced cross-presentation of tumor antigens by dermal dendritic cells (DC) in the draining lymph node (LN), driving $CD8^+$ effector T cell responses[11]. DC–TRM interactions are therefore critical for melanoma

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control and systemic immunity. Tumor-resident subsets have also been identified within regional lymph nodes (RLNs) in a pre-clinical metastatic model, providing protection against melanoma, and their presence in human melanoma involved LNs correlates with improved prognosis[12]. In cutaneous melanoma, T cell tissue residency in the TME is defined by markers such as CD103 and CD69[13], enabling rapid and localized immune responses[10]. Studies have also shown a positive correlation between the presence of tumor-resident T cells in melanoma and increased ICI clinical efficacy[3,4]. Long-term ICI responders have shared T-cell clones between the tumor and skin tumor-resident cells, but also with T effector memory in peripheral blood[14]. In addition to $CD8^+$ TRM cells, $CD4^+$ TRM can also exhibit potent cytotoxic activity through granzyme and perforin expression, directly contributing to tumor cell killing and sustaining local anti-tumor immunity, as recently demonstrated in melanoma and other solid tumors[15,16].

This supports the concept that TRM cells play an important role in local and systemic immunity. However, how TRM-like cells in humans exert anti-cancer effects under ICI therapies is unknown. The differentiation spectrum of TRM-like cells, their spatial organization, and their interactions with other immune and tumor cells are poorly understood. Moreover, how melanoma cells influence TRM-like cell surveillance within lymph nodes is unknown.

To address these questions, we profiled the immune landscape of melanoma LN metastases using bulk- and single-cell techniques, focusing our study on T cells in the context of the TME. Fresh surgical and formalin-fixed paraffin-embedded (FFPE) samples were collected from untreated, ICI-resistant, and ICI-responder patients with stage III/IV melanoma. In this work, we refer to T cells with a lymph node residency signature as '*tumor-resident T cells (TR)*', to distinguish them from circulating effector cells. TR cell ($CD69^+CD103^+$) abundance and localization within the TME of lymph node metastases were analyzed by high-dimensional flow cytometry and multiplex immunohistochemistry (mIHC), while differentiation states and T-cell receptor (TCR) clonotypes were characterized by single-cell transcriptomics. Using CellPhoneDB[17] and our SPIAT algorithm[18], we mapped intercellular interactions that support TR cytotoxicity in ICI-responder patients. Collectively, our data revealed that TR abundance and differentiation state are defining features of ICI clinical response. In addition, we identified organized "immune triads" in responder patients, where $CD8^+$ TR cells colocalize with DCs and $CD4^+$ TR cells. These triads are associated with enhanced cytotoxic programs and may support effective anti-melanoma immunity during ICI treatment.

Results

Tumor immune microenvironment of melanoma lymph node metastases reveals cytotoxic, activation, and TRM signatures are enriched in ICI responders

Lymph nodes are the central site for immune priming of T cell responses to melanoma[19]. Melanoma also establishes a pre-metastatic niche in the sentinel lymph node, disrupting its architecture[20] and the cell-cell signaling necessary for T cell differentiation[21,22] and ICI response[23]. Lymph nodes are a common site of metastasis in melanoma[24], and readily accessible for surgical sampling during treatment, providing a unique opportunity for tumor immune profiling. We analyzed tumor samples from 32 patients with stage III/IV melanoma who were untreated ($n = 13$), resistant ($n = 13$) or responsive ($n = 6$) to ICI treatment (Fig. 1a). Clinical, histologic, and treatment details are shown in Supplementary Dataset 1. Median time from ICI initiation to lymph node resection was 12 months (range: 4-60). 'ICI-resistant' tumors were collected from progressive loco-regional lymph node metastases with or without prior systemic response. 'ICI-responsive' tumors were resected as residual lymph nodes showing either FDG (fluorodeoxyglucose) uptake with histologic treatment effect or persistence during partial response (RECIST - Response Evaluation Criteria in Solid Tumors) (Supplementary Dataset 1).

To characterize the tumor microenvironment (TME), we used orthogonal profiling platforms including Nanostring, bulk RNA-seq, scRNA-/scTCR-seq, flow cytometry, and multiplex immunohistochemistry (mIHC) (Fig. 1a). The distribution of samples across these modalities was depicted in Fig. 1b, with the majority undergoing scRNA-/scTCR-seq and additional subsets profiled by two or more complementary analyses. This integrative strategy enabled cross-validation of findings and improved resolution of cellular states and spatial interactions within metastatic lymph nodes.

We hypothesized that specific accessory cells and tissue cues must be present in the melanoma TME to facilitate the ICI response. To identify immune programs underlying response, we first performed bulk RNA sequencing and Nanostring profiling, followed by gene module scoring (z-score) and gene ontology (GO) enrichment analysis (Fig. 1). ICI-responsive tumors showed significantly higher expression of genes relating to T cell cytotoxicity (*GZMH, GZMK, CST7*), activation (*CD86, IRF5, PIM1*), and tissue residency (*CRTAM, CCL3/CCL4, CXCL16*) compared to ICI-resistant tumors (Fig. 1c). A similar pattern was also observed when responders were compared with untreated cases (Fig. S1a), suggesting that these gene programs characterize effective ICI responses. In contrast, resistant versus untreated tumors showed limited differentially expressed genes (Fig. S1b), consistent with a less activated immune landscape.

We then examined immune cell-associated pathways within the TME by applying gene modules to the bulk RNA-seq data (Supplementary Dataset 2). These included IFN-γ, cDC1/cDC2, and antigen presentation signatures (Fig. 1d, e), all of which are known to support cytotoxic T cell activation (summarized in ref. 25). We observed broad upregulation of all gene modules in ICI-responsive tumors (Fig. 1f, g). In contrast, ICI-resistant and untreated tumors showed heterogeneous and incomplete activation. Cytotoxic $CD8^+$ T cells, conventional DC type 2 (cDC2), and interferon-producing M1 tumor-associated macrophages (IFN TAM M1-Like) were significantly enriched in responders (Fig. 1f). Tumor Inflammation Signature (TIS), antigen presentation and activation marker signatures were also significantly higher in ICI-responsive tumors (Fig. 1g). Gene set enrichment analysis (GSEA) confirmed that T cell proliferation, activation and cytotoxicity associated genes, dendritic cell migration, and natural killer activation were characteristic features of ICI-responsive tumors (Fig. 1h). Conversely, resistant patients exhibited upregulation of melanoma relapse and metastasis-associated pathways (Fig. 1i). A melanoma progression gene module was significantly lower in responder tumors compared to untreated and resistant tumors (Fig. S1c, d), consistent with reduced tumor cell activity and enhanced immune activation in the ICI-responsive TME.

Given our focus on TR cells, we next assessed a curated TRM signature in our cohort via Nanostring assay (Fig. 1j, Supplementary Table 1). TRM-associated genes were markedly upregulated in ICI-responsive tumors compared to resistant or untreated cases (Fig. 1k, l).

Collectively, these data demonstrate that ICI-responsive tumors are defined by high infiltration of TR cells within an immune-activated TME signature, suggesting that TR differentiation and persistence are sustained by supportive cues from accessory immune populations.

Single-cell analysis reveals distinct TME cues in response to ICI

Given this coordinated upregulation, we hypothesized that the TME contains distinct accessory-cell networks that promote TR infiltration, differentiation, or persistence. To define these networks at single-cell resolution, we performed high-resolution single-cell analysis using the Chromium 10x platform on melanoma lymph node metastases (Fig. 2a). Unsupervised clustering identified nine distinct cell populations within the TME, including T cells, B cells, myeloid cells and DCs, melanoma cells, fibroblasts, plasma cells, and endothelial cells (Fig. 2a, b and Fig. S2a). This cellular diversity provided a foundation to explore how individual immune cell states contribute to ICI responsiveness.

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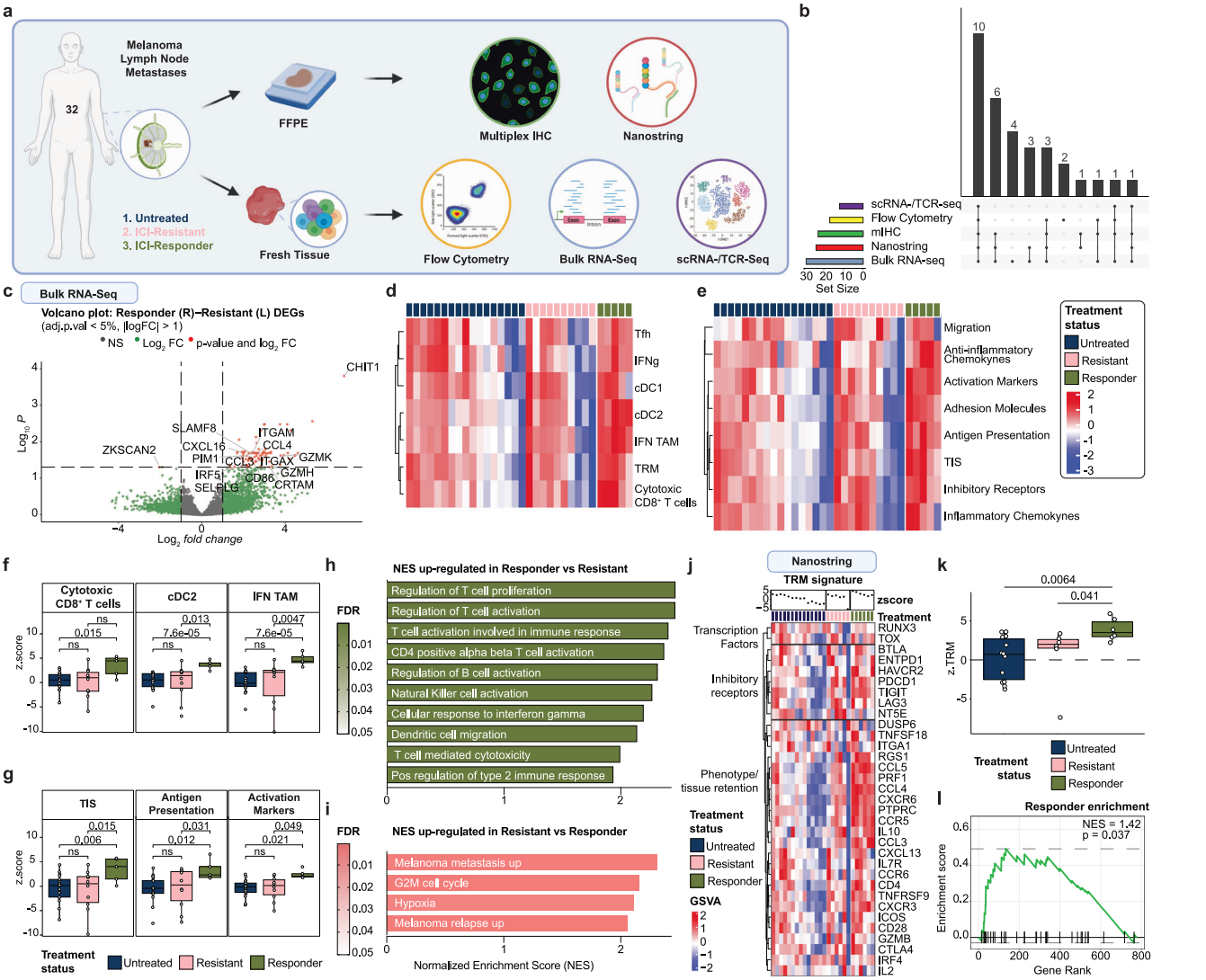


Fig. 1 | Immune activation and TRM-associated programs distinguish ICI-responsive melanoma tumors. Bulk RNA-sequencing and Nanostring techniques were adopted to analyze the TME in the melanoma patient cohort. **a** Schematics depicting the study design, patient cohorts and experimental strategy. Created in BioRender: https://BioRender.com/uh5drza. **b** UpSet plot showing the patient samples profiled by each technique. **c** Volcano plots depicting differential gene expression (DEGs) between responder (right) vs resistant (left). DEG analysis was performed using *limma* moderated t-tests (two-sided). Statistically significant genes fall at the top of the plot above the horizontal lines (p-values < 0.05). Genes were colored in red if the resulting p-value was below the given threshold based on log2 (fold-change) differential gene expression. Heatmap of significantly differentially expressed **d** immune cell and **e** cellular pathway gene modules between untreated (blue squares), resistant (pink squares) and responder (green squares) groups. Gene modules were applied to the bulk RNA-seq dataset. Box plots depicting the statistically significant z-scores of gene modules for **f** immune cells or **g** tumor inflammation and cellular pathways. p-values were calculated using two-sided Wilcoxon rank-sum tests. Box plots show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to 1.5 × the interquartile range. Individual points represent individual samples. Top significant enriched gene sets (GSEA) represented as bar plots in responder (**h**) or resistant (**i**) groups. p-values were adjusted for the False Discovery Rate (FDR). **j** Heatmap showing the expression of a curated TRM gene signature (rows) profiled by Nanostring in tumor tissue across treatment groups (columns). **k** Box plot showing the TRM signature, calculated as z-score. p-values were calculated using two-sided Wilcoxon rank-sum tests. Box plots show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to 1.5 × the interquartile range. Individual points represent individual samples. **l** Custom GSEA plot showing the TRM signature in responders compared to resistant groups. GSEA was performed using a permutation-based enrichment test. The normalized enrichment score (NES) reflects gene set enrichment in responders relative to resistant tumors, and p-values were calculated using two-sided permutation testing.

DCs are not only important for the generation and maintenance of T cell responses by cross-presenting tumor antigens, but also by mediating TR cell activation and survival through the production of cytokines including IL-12, IL-15, and TGF-β[26]. We identified DC clusters of cDC1, cDC2, DC3 and plasmacytoid dendritic cells (pDC) (Fig. 2c), with distinct transcriptional profiles (Fig. 2d). cDC1 and cDC2 in ICI-responsive tumors expressed a trend for higher levels of the antigen presentation machinery signature score compared to ICI-resistant and untreated tumors (Fig. 2e and Fig. S2b). Prior studies have shown $CD88^{-}CD1c^{+}CD163^{+}$ DC3 drove the activation and differentiation of naïve $CD8^{+}$ T cells to $CD103^{+}$ TRM in a TGF-β-dependent mechanism[27]. In our study, the DC3 subset appeared more prevalent in ICI-responsive tumors (Fig. 2f), suggesting that enhanced DC3 activity may contribute to more effective anti-tumor immune responses.

B cells modulate T cells by presenting antigens, providing co-stimulation, and producing pro-inflammatory cytokines[28]. In melanoma, B cells can potentiate ICI response[29]. In our data, B cell subsets were clustered based on gene expression profiles (Fig. 2g) and showed

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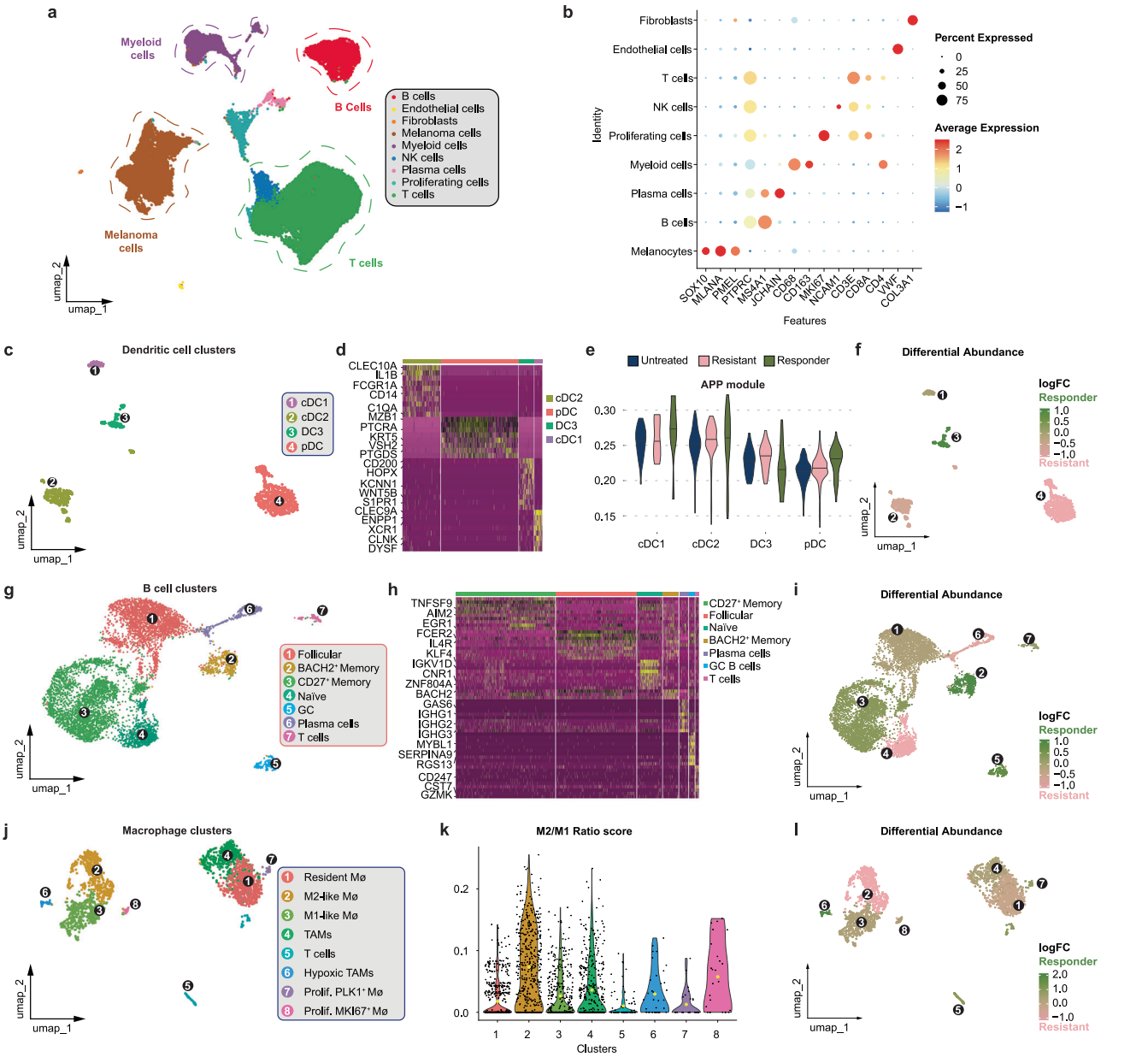


Fig. 2 | ICI-responsive tumors are enriched for pro-inflammatory dendritic cells, B cell, and macrophage states that support anti-tumor immunity. ScRNA-seq was utilized to dissect the heterogeneity of cell populations within the melanoma microenvironment. **a** Global UMAP showing cells annotated from the scRNA-seq, including melanoma cells (brown circle), T cells (green circle), B cells (red circle), and myeloid cells and DCs (pink circle). **b** Dot plot showing the expression of key marker genes across the cell types. The size of each dot represents the percentage of cells expressing the gene, while the color intensity indicates the average expression level. **c** UMAP plot illustrating DC clusters, annotated by their lineage markers and functions. **d** Heatmap of the top-10 DEGs across DC clusters. **e** Violin plot comparing the expression of key functional genes associated with antigen-presentation process (APP signature) in DC subsets among different treatment groups. A two-sided Wilcoxon signed-rank test was performed for this analysis, with no changes between groups reaching significance. **f** UMAP showing the Log fold change (LogFC) in DC cluster enrichment between responder and resistant groups. **g** UMAP plot depicting distinct B cell populations from scRNA sequencing data. The cells are grouped into clusters based on their gene expression profiles, with each cluster represented by a different color: (1) Follicular B cells, (2) BACH2$^+$ Memory B cells, (3) CD27$^+$ Memory B cells, (4) Naïve B cells, (5) Germinal Center (GC) B cells and (6) Plasma cells. **h** Heatmap displaying the top-10 gene expression profiles across different B cell clusters, showing the expression of key genes that define each cluster. **i** UMAP showing the Log fold change (LogFC) in B cell cluster enrichment between responder and resistant groups. Positive values (in green) indicate enrichment in responders, while negative values (in pink) indicate enrichment in resistant individuals. **j** UMAP plot displaying distinct macrophage clusters, each identified with specific markers, including subsets with pro-inflammatory and anti-inflammatory profiles. **k** Violin plot showing M2/M1 ratio score across macrophage clusters, with higher scores indicating an M2-like, pro-tumorigenic phenotype. **l** UMAP showing the Log fold change (LogFC) in macrophage cluster enrichment between responder and resistant groups.

Naïve B cells (*IGKV1D-8, FCRL2*), Follicular B cells (*CD200, BCL6, IL4R*), BACH2$^+$ Memory B cells (*BACH2*), CD27$^+$ Memory B cells (*CD27*), Germinal Center (GC) B cells (*CAMK1, LMO2, SERPINA9*) and plasma cells (*CD38, PRDM1, XBP1*) (Fig. 2h). The relative proportion of B cells varied across samples, with untreated and ICI-responsive tumors showing a trend toward higher overall abundance of B cells compared to resistant cases (Fig. S2a, c). Although not statistically significant, ICI-responsive tumors tended to display greater representation of BACH2$^+$ memory and germinal center B cells; whereas, ICI-resistant tumors showed higher proportions of naïve B cells and plasma cells (Fig. 2i).

These trends suggest that the presence of functionally mature B cell states may favor ICI responsiveness, likely due to their ability to support effective T cell priming, antigen presentation, and co-stimulatory signaling[30].

Tumor-associated macrophages (TAMs) exhibiting an M1-like phenotype are regarded as anti-tumor. M2-like macrophages, upon chronic exposure to INF-γ, can adopt pro-tumorigenic functions, including upregulation of PD-L1 leading to T cell suppression[31]. In our dataset, macrophage clusters displayed a diverse spectrum of phenotypes, including resident, M1-like, M2-like, and proliferating subsets (Fig. 2j). The overall proportion of macrophage subsets was different across samples (Fig. S2d), with the M2-like population showing the highest M2/M1 ratio scores (Fig. 2k). Although these trends did not reach statistical significance, ICI-responsive tumors tended to have greater representation of inflammatory M1-like macrophages, whereas ICI-resistant tumors showed a relative predominance of M2-like macrophages (Fig. 2l). Collectively, these observations suggest that ICI resistance is associated with a shift toward immunosuppressive, M2-polarized macrophage states within the TME, potentially contributing to reduced cytotoxic T cell activity and impaired therapeutic efficacy.

Together, these analyses suggest that ICI response in melanoma is associated with a coordinated immune microenvironment characterized by trends toward increased frequencies of functionally mature B cells, pro-inflammatory M1-like macrophages, and DC3 subset, which may collectively support effective T cell–mediated responses to ICI therapy.

Interferon-associated pathways in melanoma cells correlate with ICI resistance

Having established immune-specific cues linked to ICI response, we next interrogated whether intrinsic transcriptional programs within melanoma cells also contributed to treatment outcomes. Melanoma cells comprised the second largest cluster identified in the global UMAP (Fig. 2a). We stratified melanoma cells by patient groups to investigate the molecular differences associated with treatment outcomes (Fig. 3a). Enrichment analysis revealed that ICI-responsive tumors upregulated pathways associated with glycolysis and G2M cell cycle progression, with a concurrent reduction in MYC target gene expression. In contrast, ICI-resistant tumors showed strong enrichment of IFN-α/γ–associated inflammatory programs, including IL6/JAK/STAT3 and TNF-α signaling (Fig. 3b). Melanoma cells in untreated patients displayed pathways indicative of tumor progression, such as hypoxia and glycolysis (Fig. 3c). These data shed light on why the inflammatory gene modules (particularly those including the IFN-γ signature[32,33]) may not always predict ICI response as a standalone biomarker. While transient IFN-γ signaling is critical for initiating anti-tumor immunity, persistent or dysregulated IFN-γ activity can paradoxically promote immune escape by inducing PD-L1, suppressing MHC expression, and driving exhaustion in effector T cells[34].

Tumor-resident CD8$^+$ cells in lymph node metastases are enriched and functionally cytotoxic in ICI-responsive tumors

Within the T cell compartment, we identified twelve CD8$^+$ T cell clusters representing three distinct CD8$^+$ T cell phenotypes: naïve, lymph node-associated effector cells, and tumor-resident T cells (Fig. 4a). Naïve and central memory (TCM) T cells were characterized by CCR7 expression, while five clusters (consisting of ANXA1$^+$, TCF7$^+$, CCL4$^+$, EGR1$^+$, and PRF1$^+$ effector) comprised the lymph node–associated effector population. Six clusters displayed a TRM-associated transcriptional signature (*ITGAE* high; *CCR7, KLF2 and S1PR1* low; Fig. 4b, c), and we hereon define these CD8$^+$ T cells as *Tumor-Resident (TR) cells*. These included distinct transcriptional states, as follows: IKZF3$^+$ TR (e.g. *IKZF3*, *NEAT1*, *SYNE2*), GZM$^+$ TR (e.g., *ANXA1, GZMH, GZMA*), CXCL13$^+$ TR (e.g., *CXCL13, CRTAM*), NKG7$^+$ TR (e.g. *NKG7*, *GZMB*, *CD27, CXCR6*), immune checkpoint (IC)^{hi} TR (e.g., *HAVCR2, LAG3, CTLA4*), and heat-shock protein (HSP)^{hi} TR (e.g., *HSPA6*, *HSPA1A, HSPA1B, HSPB1*) (Fig. S3a).

Enrichment analysis of the twelve clusters revealed that ICI-responsive tumors were enriched with EGR1$^+$ effector and GZM$^+$ TR subsets (Fig. S3b). This was also confirmed by flow cytometry data showing that antigen-experienced CD39$^+$CD69$^+$CD103$^+$ CD8$^+$ T cells were enriched in ICI-responsive tumors (Fig. S3c).

TR clusters were also the dominant producers of IFNG (Fig. 4c, right panel). To assess functional potential, we applied a curated cytotoxic gene module (Supplementary Table 2), which confirmed that TR subsets displayed the strongest cytotoxic signatures among all CD8$^+$ T cell states (Fig. 4d). These transcriptomic findings were supported by flow cytometry. Antigen-experienced CD8$^+$ TR cells (CD103$^+$CD69$^+$CD39$^+$)[35,36] expressing GZMB were significantly enriched in ICI-responsive tumors compared to resistant or untreated cases (Fig. 4e, f). Functional assays further demonstrated robust degranulation activity (CD107a$^+$) in TR cells from responder patients following co-culture with autologous melanoma cells (Fig. 4g). Notably, this response occurred without exogenous IL-2 or any other co-stimulatory molecule, indicating that TR cells in ICI-responsive tumors are antigen-specific and capable of direct melanoma recognition. In contrast, TR cells from resistant tumors showed minimal degranulation, even upon IL-2 stimulation (Fig. 4g).

Given this strong concordance between scRNA-seq and functional validation, we hereafter define CD103$^+$CD69$^+$ cells as bona fide 'TR' T cells. Collectively, these results demonstrate that TR cells are not only enriched in ICI-responsive tumors but are also functionally cytotoxic within the TME.

Next, we asked how the transcriptional states of TR cells, particularly in relation to exhaustion status, are associated with ICI response or resistance. To this end, we applied an exhaustion[37] and a progenitor-exhausted gene modules[38,39] to the twelve CD8$^+$ T clusters (Supplementary Table 2, Fig. S3d). When we analyzed for relative mean expression of the exhaustion gene module according to ICI treatment outcomes, we found significantly lower expression of the exhaustion gene module within the IC^{hi} TR cluster in ICI-responsive tumors compared to untreated tumors (Fig. 4h). Overall, this suggests that the functional capacity of TR_{Exh} cells is better preserved in ICI-responsive tumors (Fig. S3f), potentially contributing to more effective immune-mediated tumor control.

In tumor-draining lymph nodes, TCF1$^+$ CD8$^+$ T cells are progenitor-exhausted T cells (T_{Pex}), which is an early memory T cell pool that can be reinvigorated by ICI treatments, leading to their migration into the TME and differentiation to tumor-specific effector T cells[40]. In the scRNA-seq data, we analyzed for relative mean expression of the progenitor-exhaustion gene module to understand the abundance and functional status of TR cells and their association with ICI treatment outcomes. We found that EGR1$^+$, IKZF3$^+$ TR and CXCL13$^+$ TR clusters displayed the highest T_{Pex} scores (Fig. S3d). However, there were no other differences in expression levels of the progenitor-exhaustion gene module across the seven clusters stratified by treatment groups (Fig. S3e, g).

Together, these findings demonstrate that CD8$^+$ TR cells are highly cytotoxic, antigen-specific, and functionally preserved in ICI-responsive tumors, whereas their exhaustion programs appear impaired in resistant cases, underscoring their central role in mediating effective anti-tumor immunity.

Clonal expansion of CD8$^+$ TR cells is characteristic of ICI-responsive tumors

Next, we performed single-cell TCR sequencing to investigate the clonal architecture underpinning CD8$^+$ T cell states and their association with ICI response. Hyperexpanded TCR clonotypes ($n > 100$) were observed in CD8$^+$ T cells from untreated and ICI-responsive tumors, whereas ICI-resistant tumors were enriched for small ($1 < n < 10$) and singleton ($n = 1$) clones (Fig. 4i). Within CD8$^+$ TR subsets, hyper-expanded clonotypes were detected in the GZM$^+$ TR (cluster 10) and

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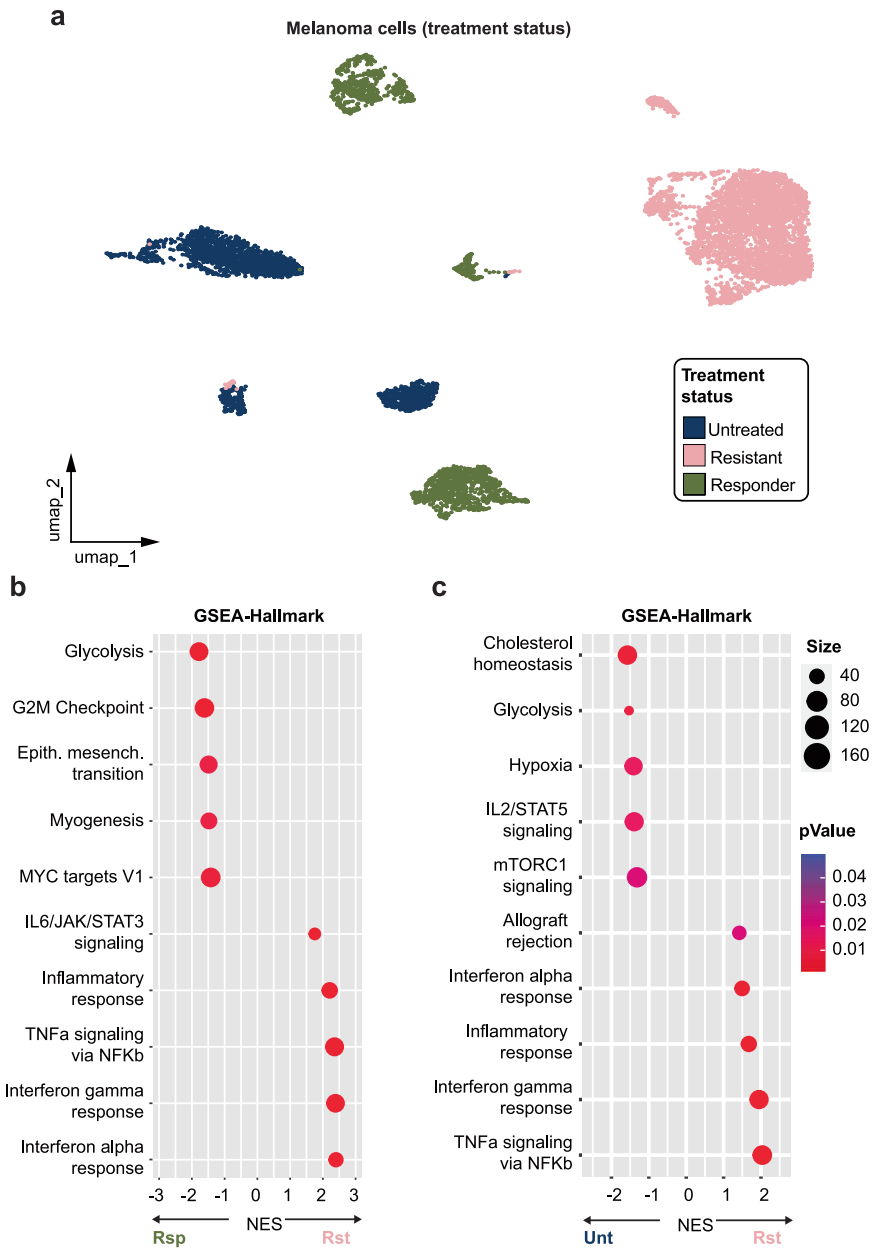


Fig. 3 | Interferon-associated pathways in melanoma cells correlate with ICI resistance. a UMAP plot showing melanoma cell distribution based on treatment status, highlighting shifts in population dynamics. **b** Gene set variation analysis (GSVA) was performed for Hallmark gene sets. Dot plot depicting pathway enrichment scores (NES) in melanoma cells, categorized by resistant (right) vs responder (left) treatment groups. **c** Dot plot summarizing key signaling pathways enriched in untreated (left) versus resistant (right) melanoma cell populations. Differences in GSVA scores between treatment groups in **b**, **c** were assessed using two-sided Student's t-tests. Dot size represents enrichment magnitude and color indicates the nominal *p*-value.

IC^{hi} (cluster 13) subsets (Fig. S4a, b). TCR expansion was particularly evident in the GZM^{+} TR cluster of responder patients (Fig. 4j) and within the IC^{hi} TR subset of untreated cases (Fig. S4c).

We next examined the clonal relationship between progenitor-like CD8^{+} TR cells (IKZF3^{+} TR and CXCL13^{+} TR clusters) and the differentiated/expanded GZM^{+} TR subset. In ICI-responsive tumors, a high proportion of unique TCR clones within IKZF3^{+} TR cells were shared and expanded with GZM^{+} TR populations (Fig. 4k and Fig. S4d). In contrast, ICI-resistant tumors exhibited limited expansion and minimal TCR sharing between IKZF3^{+} TR and GZM^{+} TR subsets (Fig. 4k and Fig. S4d), indicating a failure of clonal expansion. Minimal sharing was observed between CXCL13^{+} TR and GZM^{+} TR clusters (Fig. 4l and Fig. S4e) across all treatment groups, suggesting that CXCL13^{+} TR cells may represent a distinct, functionally specialized subset rather than a precursor population. In untreated patients, hyperexpanded clones were shared between IKZF3^{+} TR and IC^{hi} TR subsets (Fig. S4f, g), indicating that these transitional states may precede the formation of terminal cytotoxic TR cells.

Collectively, these data demonstrate that successful ICI responses are characterized by robust clonal expansion and transition from progenitor-like IKZF3^{+} TR cells to differentiated cytotoxic TR states, whereas ICI resistance is marked by impaired clonal amplification and diversification.

Cytotoxic CD4^{+} tumor-resident cells contribute to ICI response

Effector functions within the CD4^{+} T cell compartment, including cytokine production and co-stimulation, are crucial for orchestrating effective anti-tumor immunity by enhancing CD8^{+} effector and tumor-

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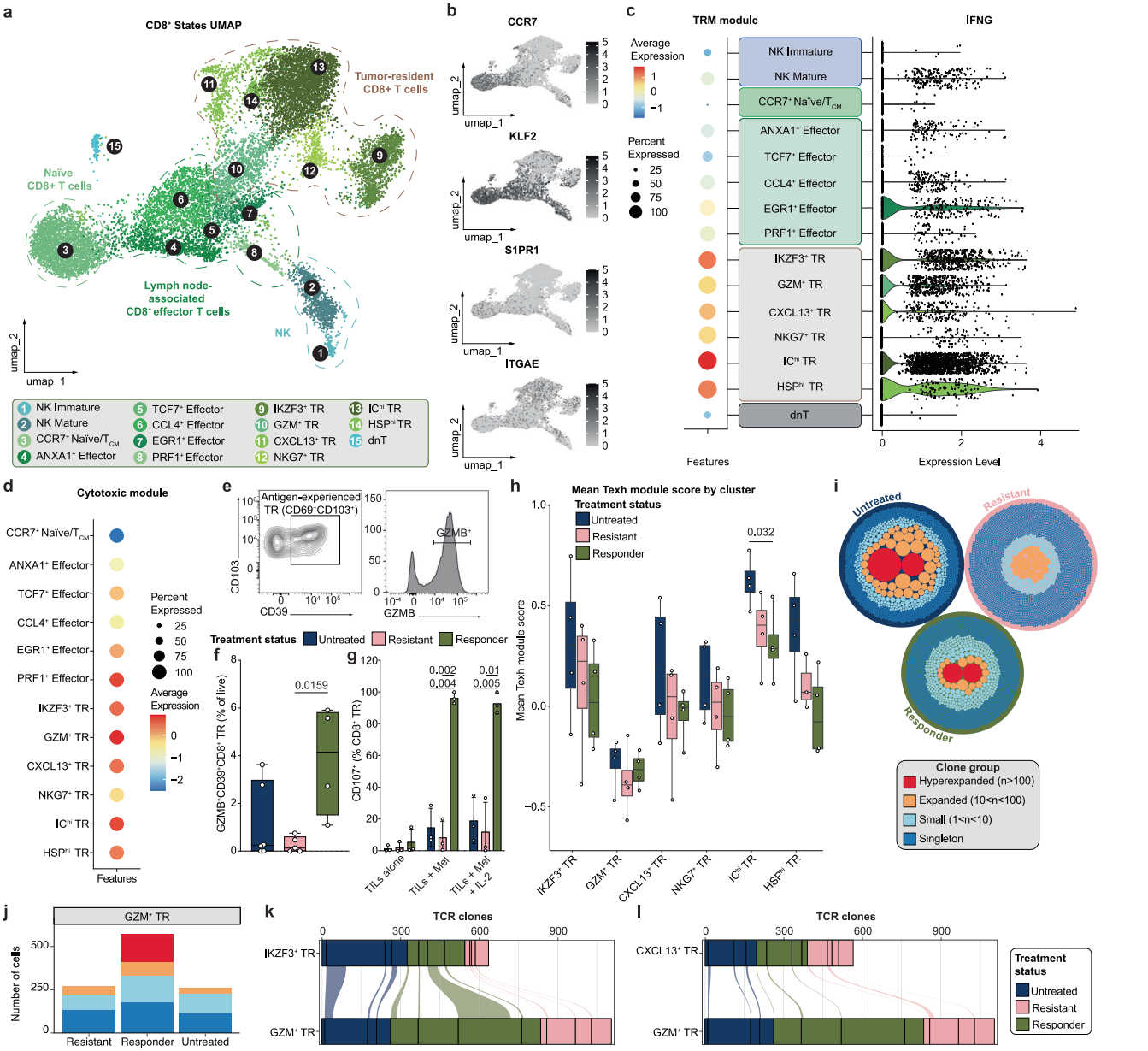


Fig. 4 | Tumor-resident CD8⁺ T cells are cytotoxic and clonally expanded in ICI-responsive tumors. scRNA-/scTCR-seq was performed to explore T cell heterogeneity in melanoma LN metastases. **a** UMAP of CD8⁺ T cell subsets annotated by cluster identity. **b** Feature plots showing expression of CCR7, KLF2, S1PR1, and ITGAE (CD103) defining TR versus lymph node–associated phenotypes. **c** Dot plot showing the expression of the TRM module across CD8⁺ T cell subsets (left). Violin plot displaying IFNG expression across CD8⁺ T cell populations (right). **d** Cytotoxic gene module expression across CD8⁺ T cell subsets. Dot plot shows average expression (color gradient) and percent expression (circle area) of selected transcripts. **e** Contour plots representing antigen-experienced TR cells (CD39⁺CD69⁺CD103⁺, left) and Granzyme B positive TR cells (GZMB⁺, right). **f** Box plot showing the frequency of CD39⁺GZMB⁺CD8⁺ TR cells in the different treatment groups. **g** Quantification of CD107a⁺ degranulation following co-culture with autologous melanoma cells, showing increased cytotoxic activity in ICI responders. Data in **f**, **g** are represented as mean ± SEM. Box plots show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to the minimum and maximum values. Individual points represent individual samples. *p*-values were calculated using two-sided Mann-Whitney test. **h** Box plot indicating the mean T cell exhaustion signature in CD8⁺ TR subsets. Box plots show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to 1.5 × the interquartile range. Individual points represent individual samples. Two-sided Wilcoxon signed-rank test was performed for this analysis. **i** Dot plots showing TCR repertoire abundance in CD8⁺ T cells from untreated, resistant or responder patients. Each dot represents a unique TCR sequence, and its area reflects clonal expansion. **j** Stacked bar plot showing TCR clones by cell type annotation across treatment groups. Clone sizes are classified as hyperexpanded ($n > 100$; red), expanded ($10 < n < 100$; orange); small ($1 < n < 10$; light blue), or singleton (blue). Sankey plots depict TCR clones shared between **k** CD8⁺IKZF3⁺ TR and CD8⁺GZM⁺ TR or **l** CD8⁺CXCL13⁺ TR and CD8⁺GZM⁺ TR populations. Clones are color-coded by treatment status: untreated (blue), resistant (pink), responders (green).

resident T cell responses in melanoma[41,42]. To delineate which CD4⁺ subsets may support CD8⁺ TR cells during ICI response, we isolated CD3⁺/CD4⁺-expressing cells from our scRNA-seq dataset and identified eleven CD4⁺ clusters representing six distinct transcriptional states: naïve (BACH2⁺ Naïve/T_{CM}), activated (CD69⁺), T follicular helper (T_{fh}) (CXCL13⁺ T_{fh}), T_{h1} (CCL5⁺, ANXA1⁺, MAF⁺ T_{h1}), TR (NKG7⁺ TR), and T_{Reg} (FOXP3⁺, IKZF2⁺, BCL2⁺, NEAT1⁺ T_{Reg}) (Fig. 5a). The CD4⁺ NGR7⁺ TR cluster exhibited the strongest TRM gene module expression (Fig. 5b), consistent with a tissue-resident phenotype and analogous to CD8⁺ TR cells. Differential abundance analysis showed that NKG7⁺ TR subsets

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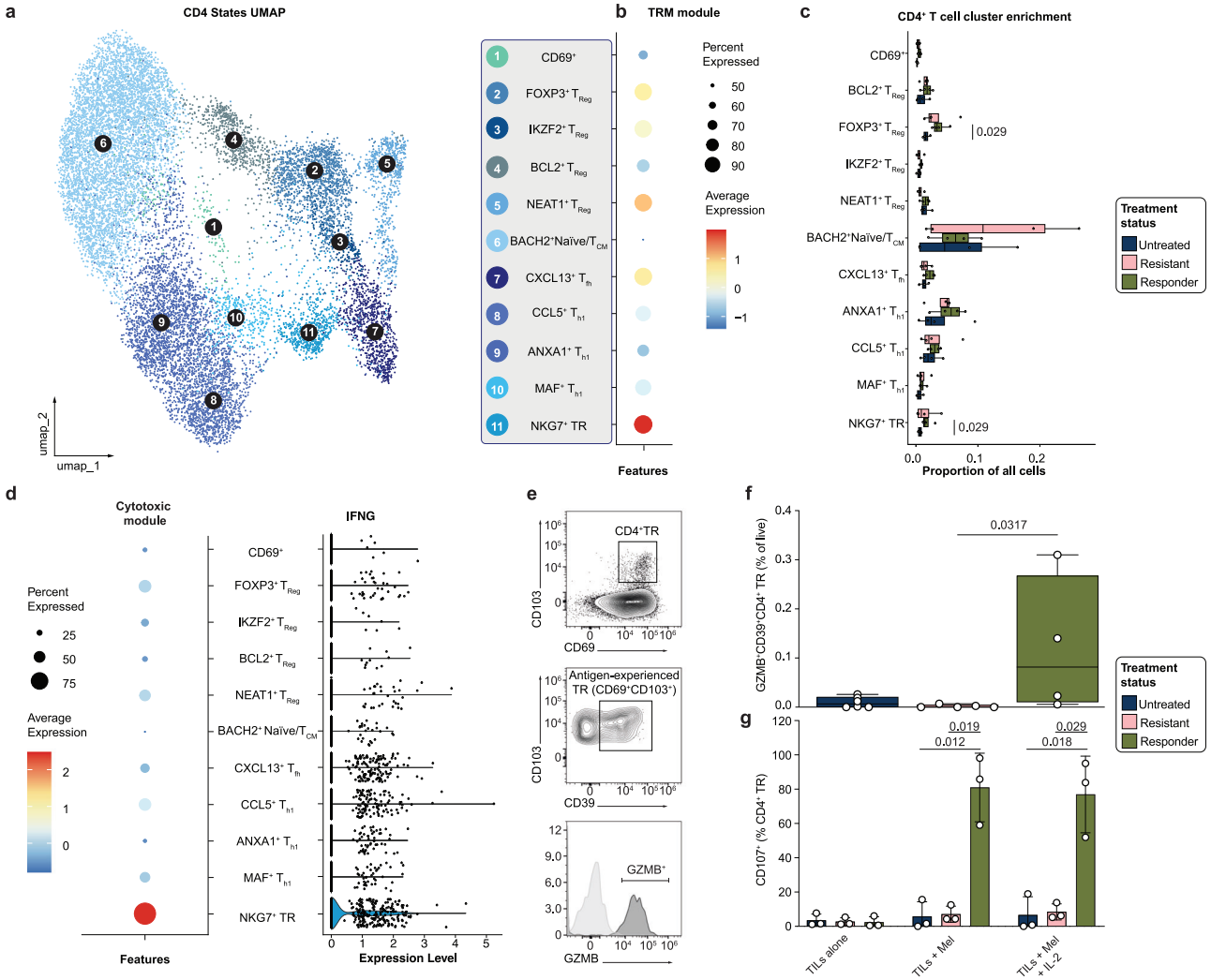


Fig. 5 | Abundance of cytotoxic CD4⁺ tumor-resident T cells is associated with ICI response. CD4⁺ T cells were extrapolated from the scRNA-seq dataset to identify specific different cellular subsets. **a** UMAP displaying the 11 CD4⁺ T cell clusters. **b** TRM gene module applied to CD4⁺ T cell subsets. Dot plot shows average expression (color gradient) and percent expression (circle area) of selected transcripts in the indicated clusters. **c** Differential analysis of CD4⁺ T cell cluster proportions between resistant (pink), responder (green) and untreated (blue) patients. Wilcoxon signed-rank test was performed for this analysis. Box plots show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to 1.5 × the interquartile range. Individual points represent individual samples. Significance of differential abundance was assessed using glmQLFTest from the *edgeR* package. **d** Cytotoxic gene module (left) expression applied to CD4⁺ T cell subsets. Dot plot shows average expression (color gradient) and percent expression (circle area) of selected transcripts in the indicated clusters. Violin plots show the expression level of the IFNG gene (right). **e** Flow cytometry analysis of CD4⁺ TR cells in the tumor microenvironment. Top: CD4⁺ TR cells were identified based on co-expression of CD69 and CD103. Middle: A subset of antigen-experienced CD4⁺ TR cells (CD39⁺) was further gated. Bottom: Histogram showing intracellular staining for Granzyme B (GZMB) in the CD4⁺ TR compartment, revealing a distinct population of GZMB⁺ CD4⁺ TR cells suggestive of cytotoxic potential. **f** Box plot of flow cytometric data showing the frequency of GZMB⁺CD39⁺CD4⁺ TR cells in the different treatment groups. Each dot represents an individual patient sample. **g** Functional CD107a degranulation assay showing increased cytotoxic activity of CD4⁺ TR cells from responder patients upon co-culture with autologous melanoma cells. Data in **f**, **g** are represented as mean ± SEM. Box plots show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to the minimum and maximum values. Individual points represent individual samples. *p*-values were calculated using nonparametric Mann-Whitney test.

were enriched in ICI-responsive tumors (Fig. 5c), also confirmed by flow cytometric analyses (Fig. S5b). The BACH2⁺ Naïve/T_{CM} subset predominated in ICI-resistant tumors (Fig. 5c and Fig. S5c). This indicates that differentiated and activated CD4⁺ states are associated with therapeutic responsiveness.

In lymphoid tissues, CD4⁺ T_{fh} cells provide essential help for CD8⁺ T cell priming and differentiation (reviewed in ref. 43). Notably, CD4⁺ TRM cells have been shown to share transcriptional and functional features with both T_{h1} and T_{fh} lineages[44]. In our dataset, the NKG7⁺ TR subset expressed 206 differentially expressed genes, including GZMA, GZMK, NKG7, and PDCD1, indicating cytotoxic potential similar to that observed in CD8⁺ TR cells (Fig. S5d). In contrast, CXCL13⁺ T_{fh} expressed 74 T_{fh} cell-associated genes (e.g., *CXCL13, PDCD1, SH2D1A, NR3C1* and *PPP1CC*) (Fig. S5d). Notably, we identified 96 overlapping expressed genes (25.5%) between these two subsets, including genes related to immune modulation (e.g., *CD27*, *CD82*, *CTLA4*, *TIGIT*), transcriptional regulation of T cell function in lymph nodes (e.g., *ARID5B, BATF, ID2*) and genes preventing apoptosis for long-term residency (e.g., *CFLAR, DUSP2, DUSP4*) (Fig. S5d, e). This overlap suggests that CD4⁺ TR cells may adopt T_{fh}-like helper functions while retaining cytotoxic capacity.

To further investigate cytotoxic potential, we applied a curated cytotoxic gene module (Supplementary Table 2) across CD4⁺ clusters. Among all subsets, only NKG7⁺ TR cells showed high expression of cytotoxic effectors and the highest IFNG levels (Fig. 5d). Flow

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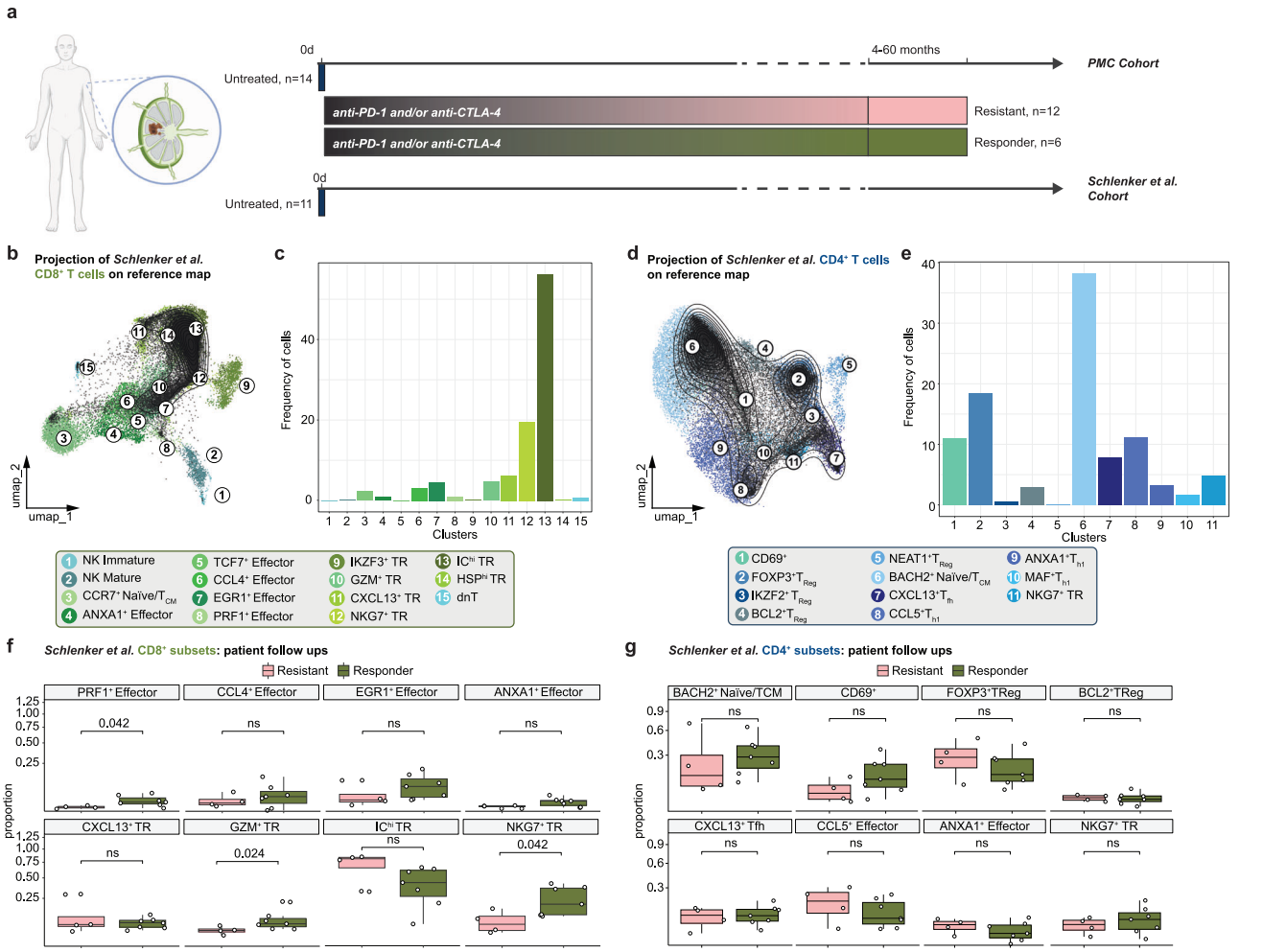


Fig. 6 | Tumor-resident CD8⁺ T cell states are reproducible across independent melanoma cohorts and associate with clinical response. a Schematic representation of patient cohorts, including untreated, ICI-treated, and responder/resistant groups across different datasets (PMC - PeterMac, Schlenker et al. cohort). Time from initiation of treatment to tumor resection is shown. d = days. Created in BioRender: https://BioRender.com/xen2qu2. **b** UMAP reference maps for CD8⁺ T cells generated from our dataset, showing projections of CD8⁺ T cell subsets from Schlenker et al. dataset across various differentiation and functional states. CD8⁺ subsets include clusters such as effector, IC^{hi} TR, and NKG7⁺ TR cells. **c** Frequency distributions of CD8⁺ T cell subsets across the Schlenker et al. cohort. **d** UMAP reference maps for CD4⁺ T cells generated from our dataset, showing projections of CD4⁺ T cell subsets from Schlenker et al. dataset across various differentiation and functional states. CD4⁺ subsets include activated T cells, T_{Reg}, T_{h1}, and TR subsets. **e** Frequency distributions of CD4⁺ T cell subsets across the Schlenker et al. cohort. **f** Proportions of CD8⁺ T cell subsets in patient follow-ups, segregated by clinical response (Resistant vs. Responder). **g** Proportions of CD4⁺ T cell subsets in patient follow-ups, segregated by clinical response (Resistant vs. Responder). Box plots in **f**, **g** show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to 1.5 × the interquartile range. Individual points represent individual samples. Significance of differential abundance in **f**, **g** was assessed using glmQLFTest from the *edgeR* package.

cytometry confirmed that antigen-experienced CD4⁺ TR cells expressed high levels of GZMB and were significantly enriched in ICI-responsive tumors compared with resistant or untreated cases (Fig. 5e, f). Moreover, CD4⁺ TR cells from responders displayed robust degranulation (CD107a⁺) after co-culture with autologous melanoma cells, even without IL-2 stimulation, indicating antigen-specific cytotoxic activity (Fig. 5g). Together, these findings confirm that cytotoxic CD4⁺ TR cells are abundant and functionally active in ICI-responsive tumors.

To assess antigen-driven clonal dynamics, we analyzed CD4⁺ TCR repertoires. TCR expansion in CD4⁺ T cells is indicative of antigen recognition through MHC class II molecules[24,25]. However, we observed limited clonal expansion with 'small clones' ($1 < n < 10$) present in ICI-responsive and untreated tumors, and ICI-resistant tumors had the highest proportion of singleton clones (Fig. S6a). When mapped to phenotypic clusters, small expanded clones were predominantly localized within CD4⁺ TR, T_{h1}, T_{fh}, and T_{Reg} subsets, whereas BACH2⁺ Naïve/T_{CM} cells contained the highest number of singleton TCRs (Fig. S6b, c).

Together, these findings suggest that CD4⁺ TR cells may contribute to ICI response through both cytotoxicity and helper-like functions that sustain CD8⁺ TR activity.

High levels of infiltrating effector CD8⁺ TR cells pre-treatment predict ICI response

To validate our findings, we analyzed both CD8⁺ and CD4⁺ T cells from melanoma metastases collected pre- and post-ICI treatment from two independent scRNA-seq datasets (Fig. 6a and Fig. S7a). CD8⁺ (Fig. 6b, c) and CD4⁺ (Fig. 6d, e) T cells were annotated using the reference maps generated in our study (Fig. 3a and Fig. 4a).

In the first validation dataset (pre-treatment samples – Schlenker et al.[45]), we examined melanoma lymph node metastases stratified into ICI-responsive and ICI-resistant groups according to reported clinical outcomes. Among CD8⁺ T cells, the four most frequent cell types were all TR subsets (Fig. 6c), with PRF1⁺ effector, GZM⁺ TR, and NKG7⁺ TR significantly enriched in ICI-responsive tumors (Fig. 6f). In contrast, there were no significant differences across CD4⁺ T cell subsets

between patient groups (Fig. 6g). Overall, these results corroborate our primary cohort and confirm that a high pre-treatment abundance of effector CD8⁺ TR cells predicts subsequent ICI responsiveness.

In the second validation dataset (Sade-Feldman et al.[23]), the tumor samples differed from ours and the first dataset (Schlenker et al.[45]) in both tissue origin (including multiple metastatic sites beyond lymph nodes) and post-treatment sampling timepoint (earlier, at ~5–9 weeks, compared with 4–60 months in our cohort; Fig. S7a). After annotating samples by ICI response, CD8⁺ T cells (Fig. S7b) were found to be dominated by the T_{Pex}-like IKZF3⁺ TR subset (~60% of cells) (Fig. S7c), consistent with the original publication[23]. This likely reflects the earlier post-ICI sampling window[23], when T_{Pex} cells are a hallmark of the early immune response, whereas residency phenotypes emerge at later timepoints. Nonetheless, within TR subsets, CD8⁺ IC^{hi} TR cell abundance was increased in the ICI-responsive tumors post-treatment (Fig. S7d). The CD4⁺ NKG7⁺ TR subset was more abundant than other CD4⁺ T cells (Fig. S7e, f), however, these were not significantly enriched in responders (Fig. S7g), suggesting context- and timing-dependent differences in TR-cell activation across tumor sites and treatment stages.

Together, these validation analyses confirm that the presence of cytotoxic, tissue-resident CD8⁺ T cells before therapy is a robust predictor of ICI response, reinforcing their potential as a biomarker for patient stratification.

Abundance of CD4+ and CD8+ TR cells within the TME characterizes ICI treatment response

It has been established that CD69⁺CD103⁺ expression on T cells identifies a subset with a tissue-residency phenotype[46]. We hypothesized that a high abundance of CD69⁺CD103⁺ T cells are necessary for an effective anti-tumor response to ICI. To investigate this, we performed multiplex immunohistochemistry (mIHC) on lymph node metastases to assess the distribution and density of CD4⁺ and CD8⁺ TR cells relative to melanoma cells and normal lymphoid structures (Fig. 7a). Using serial sections and mark-up of hematoxylin and eosin (H&E)–stained slides, tumor and normal lymphoid regions were delineated for each sample (Fig. 7b and Fig. S8a). Corresponding mIHC panels included markers for SOX10 (melanoma), CD3, CD4, CD8, CD69, CD103, and TCF1 (Fig. 7c, d; Fig. S9a, b; Supplementary Table 3). These combinations identified four TR phenotypes per T cell lineage: TCF1⁺CD4⁺ TR, TCF1⁻CD4⁺ TR, TCF1⁺CD8⁺ TR, and TCF1⁻CD8⁺ TR (Fig. 7c).

Analysis of total CD4⁺ and CD8⁺ T cell densities across both normal lymphoid tissue and tumor regions revealed that while CD4⁺ T cell abundance did not differ significantly between treatment groups, CD8⁺ T cells were markedly enriched in the tumor areas of ICI-responsive cases ($p = 0.009$) (Fig. S9c). In contrast, within normal lymphoid tissue, CD8⁺ T cell frequencies were higher in untreated and resistant tumors compared with responders ($p = 0.013$ and $p = 0.036$, respectively), suggesting that ICI response is associated with selective intratumoral accumulation of CD8⁺ T cells rather than global lymphoid expansion.

Quantitative image analysis revealed that both TCF1⁺ and TCF1⁻ CD8⁺ TR subsets were significantly enriched within tumor regions of ICI-responsive lymph nodes compared with resistant or untreated cases (Fig. 7e). In contrast, CD8⁺ TR frequencies in normal lymphoid regions were comparable across all treatment groups. For CD4⁺ TR cells, a similar trend was observed, with TCF1⁺ CD4⁺ TR and TCF1⁻ CD4⁺ TR cells significantly increased in tumors from ICI-responsive patients (Fig. 7f). Whole-slide spatial maps (Fig. S8b) confirmed that these CD4⁺ and CD8⁺ TR populations preferentially localized within tumor nests rather than surrounding lymphoid tissue, particularly in ICI responders.

Consistent with these findings, extended phenotypic analyses incorporating additional CD4⁺ and CD8⁺ subsets revealed a broader enrichment of effector-like populations in ICI-responsive tumors (Fig. S9c, d). Apart from CD8⁺ TR cells, the CD69⁺CD8⁺ subset exhibited significantly higher densities in responder tumor regions compared to resistant or untreated cases. Among CD4⁺ T cells, CD69⁺CD4⁺ cells followed the same pattern of enrichment. In contrast, within normal lymphoid areas, no consistent treatment-related differences were observed, highlighting that the selective expansion of activated and tissue-resident T cells is a tumor-restricted feature of ICI responsiveness.

Collectively, these findings demonstrate that intratumoral enrichment of both CD4⁺ and CD8⁺ TR cells, especially their TCF1⁺ progenitor-like subsets, is a defining feature of ICI-responsive melanoma metastases.

CD8⁺ TR cells, CD4⁺ TR cells, and DCs form an in situ immune triad for ICI response

To understand how cell–cell interactions and spatial organization of tumor-resident T cells contribute to ICI response, we performed neighborhood clustering using DBScan on multiplex IHC datasets (Fig. 8a). Cells within a 30 µm radius were grouped into spatial neighborhoods, and compositionally similar neighborhoods were classified into archetypes. Differential abundance analysis across treatment groups identified eight discrete archetypes with distinct immune cell compositions (Fig. 8b, c). Among these, "CD4⁺ TR-dominated", "mixed CD8⁺ TCF1⁻ TR", and "mixed CD8⁺ CD69⁺" archetypes were significantly enriched in ICI-responsive tumors ($p = 0.038$, $p = 0.0003$), whereas the "CD4⁺ TCF1⁺ naïve-like" archetype predominated in resistant cases ($p = 0.001$). These findings indicate that ICI response is associated with spatial clustering of activated TR cells within pro-inflammatory neighborhoods. To confirm the robustness of these findings, we performed a permutation analysis comparing archetype composition across treatment groups, which confirmed strong reproducibility of the TR-dominated and naïve-dominated neighborhood signatures (Fig. S10a). When the same analysis was applied to normal lymphoid regions, the overall archetype structure remained conserved, but CD4⁺ TR-dominated neighborhoods were significantly enriched only in ICI-responsive tumors ($p = 0.00002$; Fig. S10b).

To explore potential molecular crosstalk underlying these spatial relationships, we analyzed ligand–receptor (L–R) interactions between T-cell states using CellPhoneDB. The interaction map revealed dense connectivity among CD8⁺ TR, CD4⁺ TR, and effector subsets in ICI-responsive tumors (Fig. 8d). Comparative analysis of responders versus resistant tumors identified selective upregulation of CCL4–CCR5 and CCL5–CCR4 axes in responder patients (Fig. 8e). Corresponding CellPhoneDB analyses in resistant and untreated tumors (Fig. S10c, d) revealed markedly reduced TR–TR and TR–effector connectivity, further supporting that the pro-inflammatory spatial architecture identified in responders reflects an active and dynamic immune microenvironment rather than baseline lymphoid organization. These pathways are consistent with enhanced T-cell recruitment, adhesion, and activation in responders, suggesting that coordinated spatial and molecular interactions between CD8⁺ and CD4⁺ TR cells establish a pro-inflammatory niche that supports ICI efficacy.

To validate whether the predicted TR–TR interactions occur in situ, we quantified the physical proximity of CD8⁺ TR and CD4⁺ TR cells within the multiplex IHC dataset. Direct spatial analysis confirmed that CD8⁺ TR–CD4⁺ TR cell interactions were significantly more frequent within the tumor regions of ICI-responsive lymph nodes compared with resistant or untreated cases (Fig. 9a). These interactions were specific to the TME, as no differences were observed within the surrounding normal lymphoid tissue, suggesting that productive TR–TR communication arises selectively in response to tumor-derived cues rather than baseline lymphoid architecture.

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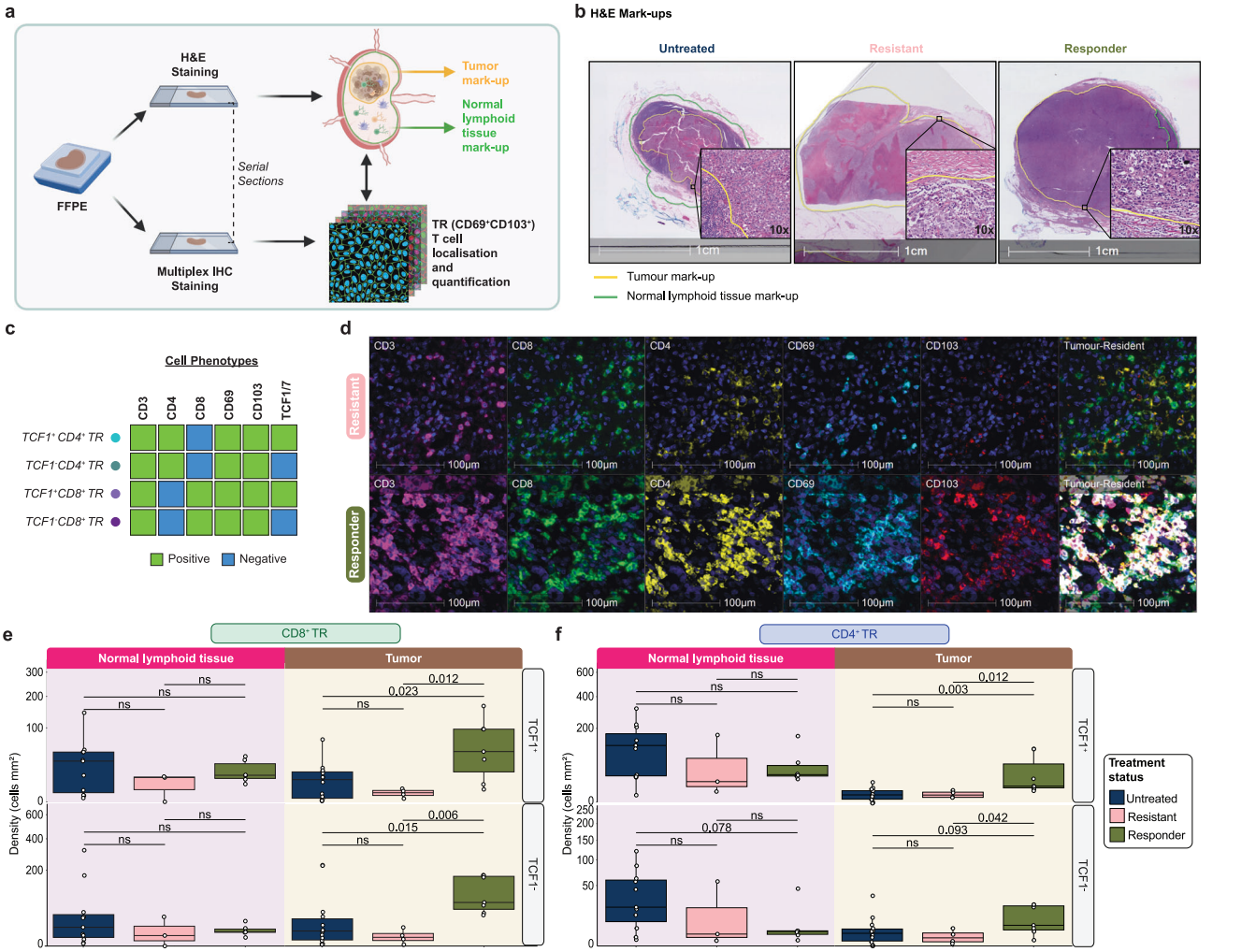


Fig. 7 | CD69+CD103+ T cell abundance in the TME is correlated with response to immune checkpoint therapy. a Workflow for spatial localization and quantification of CD69$^+$CD103$^+$ T cells. Formalin-fixed paraffin-embedded (FFPE) sections were stained with hematoxylin and eosin (H&E) and a pathologist assigned tumor and stromal regions. Multiplex immunohistochemistry (mIHC) was used to identify and localize CD69$^+$CD103$^+$ cells within the tumor and normal tissues. Created in BioRender: https://BioRender.com/4sf95m8. **b** Representative H&E-stained lymph node sections from untreated, resistant, and responder melanoma patients. Tumor regions (yellow outlines) and adjacent non–tumor-involved lymphoid tissue (green outlines) were annotated by a pathologist and used to guide downstream spatial and molecular analyses. 10X zoomed insets shows critical areas of interest at the border between tumor areas and normal lymphoid tissue. Images are representative of independent samples from untreated ($n = 13$), resistant ($n = 4$), and responder ($n = 6$) patients. **c** mIHC markers (CD3, CD4, CD8, CD69, CD103, TCF1) used to identify different TR cell phenotypes. **d** mIHC composite image of a representative tissue showing the stained markers to identify the different T cell populations in FFPE tissues. Images are representative of independent samples from resistant ($n = 4$) and responder ($n = 6$) patients. Cell density analyses of CD8$^+$ (**e**) and CD4$^+$ (**f**) CD69$^+$CD103$^+$ T cells performed with the SPIAT toolkit. The bar plots show the abundance of cells per square millimeter (mm2) in both normal and tumor areas. Box plots show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to 1.5 × the interquartile range. Individual points represent individual samples. Statistical significance was assessed by two-side Wilcoxon signed-rank test. SPIAT Spatial Image Analysis of Tissues, mIHC multiplex immunohistochemistry.

Next, we assessed the spatial distribution of individual T-cell subsets relative to melanoma cells. Both CD8$^+$ TR and CD4$^+$ TR populations, particularly the TCF1$^-$ cytotoxic subsets, were located in closer proximity to melanoma cells in ICI-responsive tumors than in resistant or untreated samples (Fig. 9b, c). In contrast, naïve-like and TCF1$^+$ precursor T-cell subsets were positioned farther from tumor nests, consistent with their less differentiated state and lower effector potential. Collectively, these findings reveal that effective ICI response is characterized by enhanced TR–TR interaction density within the tumor core and by preferential localization of cytotoxic TR cells near melanoma cells, supporting direct tumor surveillance and effector engagement within the tissue microenvironment.

Spatial organization of immune cells within tumor niches has emerged as a critical determinant of effective anti-tumor immunity, with recent studies demonstrating that higher-order cellular architectures can define response to immunotherapy[47]. Building on the observed enrichment of TR–TR neighborhoods, we next examined whether CD8$^+$ TR, CD4$^+$ TR, and DCs physically interact within the TME of ICI responders. Given the enrichment of DC-related signatures and antigen-presentation programs in responders across bulk and single-cell datasets (Fig. 1g and Fig. 2e), we asked whether DCs were positioned in close proximity to CD8$^+$ and CD4$^+$ TR cells within tumor regions of ICI-responsive lymph nodes. Multiplex IHC revealed that CD8$^+$ and CD4$^+$ TR populations were frequently enriched within tumor nests of responders (Fig. 7e, f, Fig. 10a, Fig. S11a), whereas overall DC density was comparable between the two treatment groups (Fig. S11b). Proximity mapping demonstrated that CD8$^+$ TR, CD4$^+$ TR, and DCs were often positioned within 30 µm of one another, forming discrete immune triads (Fig. 10b, c). Quantification revealed that DC-TR triad density was significantly higher in tumor regions of responders

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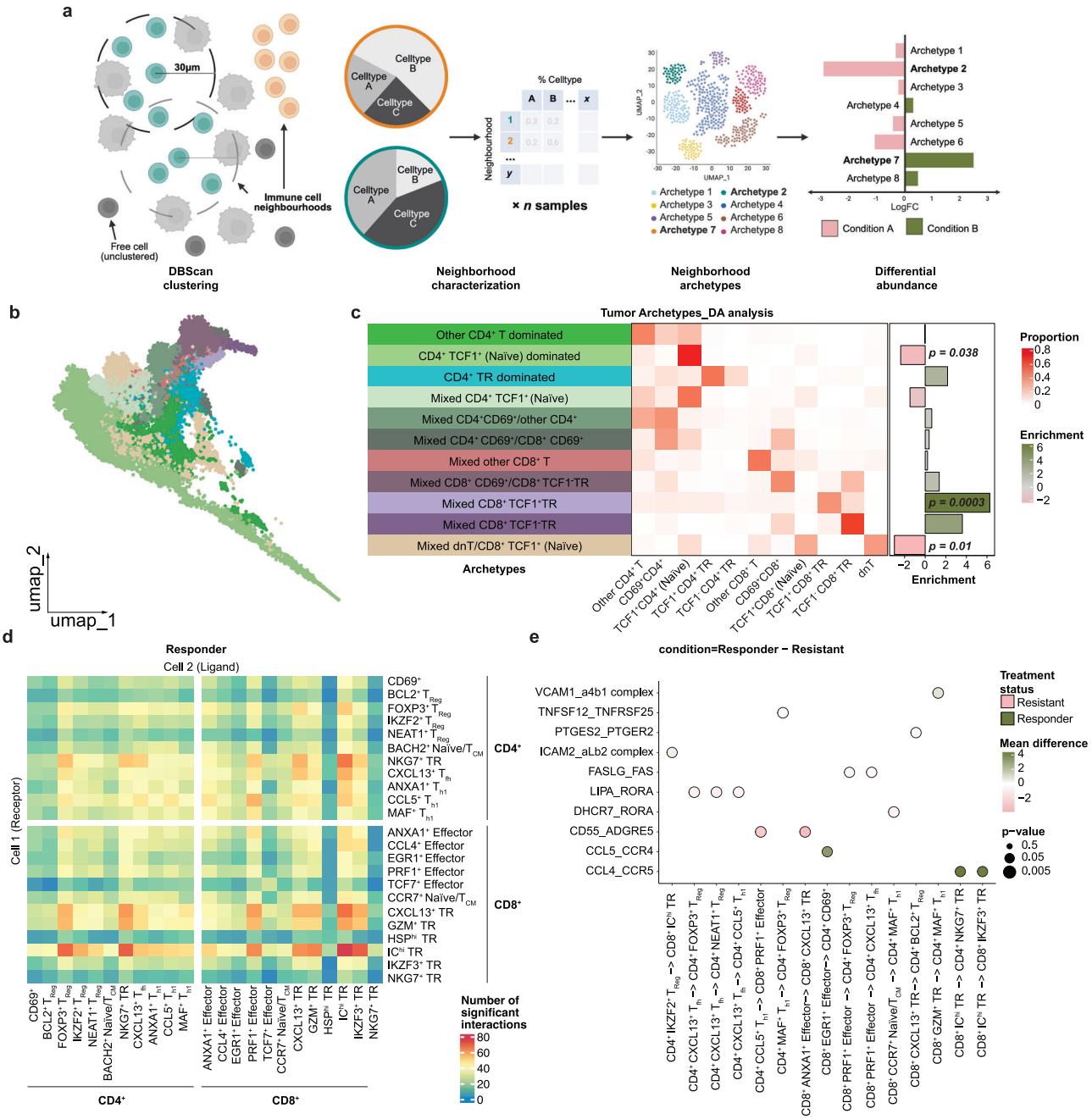


Fig. 8 | Cell archetypes including TCF1-CD8+ TR cells are enriched in ICI-responsive tumors. SPIAT tool and CellPhoneDB package were used to analyze cell-to-cell interaction, neighborhood characterization and archetypes. **a** Workflow of the *SPIAT* analysis, including *DBScan* clustering to identify immune cell neighborhoods, neighborhood characterization, and identification of neighborhood archetypes, with differential abundance analysis indicating significant archetypes across conditions. Created in BioRender: https://BioRender.com/wrxslc0. **b** UMAP projection depicting the clustering of T cell archetypes. Each point represents a characterized neighborhood, color-coded by their assigned archetype. Archetypes were defined based on the cellular composition of each neighborhood and are labeled in the legend. Distinct clusters correspond to archetypes such as *Other CD4⁺ T Dominated, CD4⁺ TR Dominated, CD4⁺ TCF1⁺ (Naïve) Dominated*, and mixed populations including *Mixed CD8⁺ TCF1⁺ TR* and *Mixed CD8⁺ TCF1⁻ TR*. **c** Differential abundance (DA) analysis of archetypes in tumor areas displaying the enrichment and proportion of various immune cell-dominated archetypes, with significance levels annotated. Differential abundance significance of archetypes was assessed using glmQLFTest from the *edgeR* package. *P*-values were derived from quasi-likelihood F-tests. **d** Heatmap depicting the number of significant interactions between T cell subtypes in ICI-responder samples as identified with CellPhoneDB. **e** Inferred ligand–receptor interactions enriched in responder versus resistant melanoma lymph node metastases. Selected ligand–receptor pairs (y-axis) and directional cell–cell interactions (x-axis) were analyzed across CD4⁺ and CD8⁺ T cell subsets using curated interaction databases. Dot size reflects statistical significance (*p*-value) calculated by permutation analysis, and color represents the direction and magnitude of mean interaction score difference between responders and resistant samples (green = enriched in responders; pink = enriched in resistant).

compared to resistant tumors ($p = 0.0436$; Fig. 10d), while no difference was observed in normal lymphoid tissue. Importantly, DC/non-TR triad density did not differ between treatment groups (Fig. S11c). Together, these findings indicate that DC–TR triad enrichment in responders reflects structured spatial organization rather than differences in overall TR and DC abundance.

To investigate the molecular basis of this interaction, we performed ligand–receptor pairing analysis between DCs and TR cells. In

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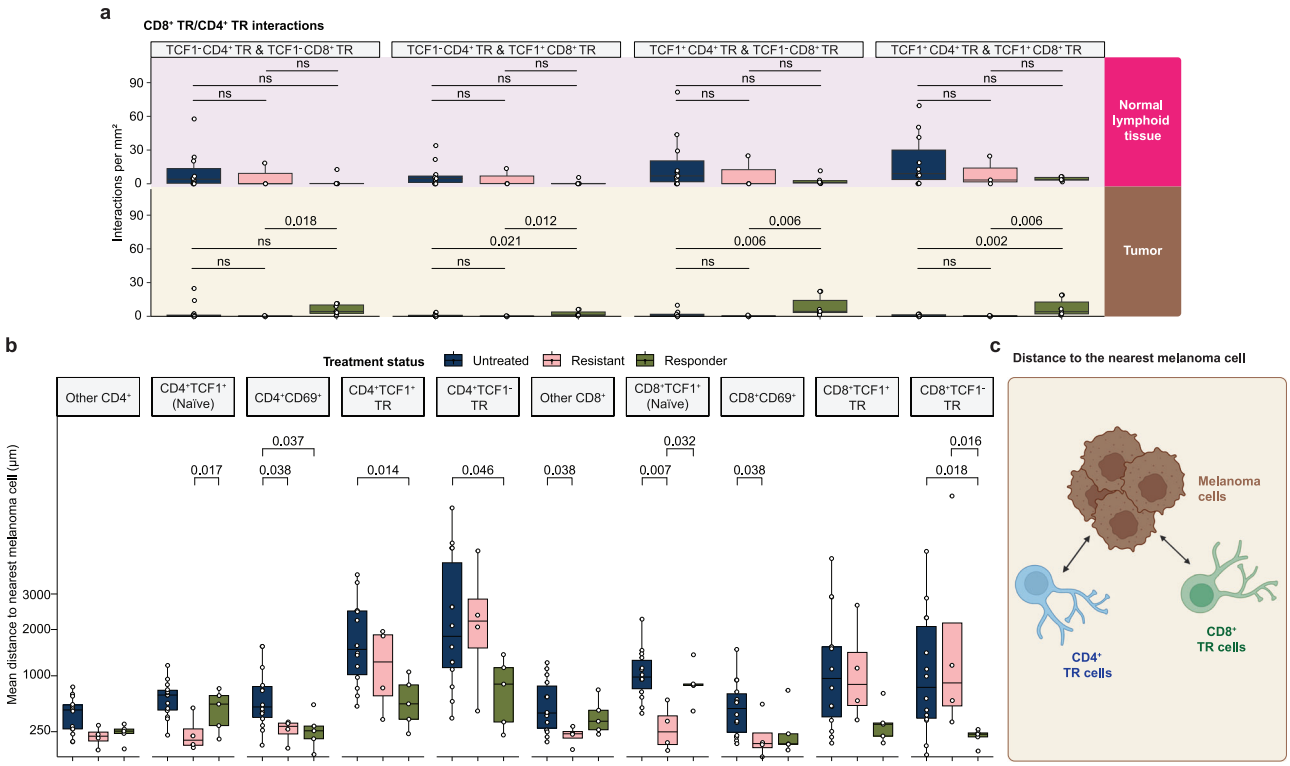


Fig. 9 | CD8⁺ and CD4⁺ tumor-resident T cells preferentially interact and localize near melanoma cells in ICI-responsive tumors. a Cell–cell interaction frequencies per mm² between CD8⁺ TR and CD4⁺ TR phenotypes stratified by tissue compartment, with normal lymphoid regions shaded purple and tumor regions shaded tan; comparisons include TCF1⁻CD4⁺ TR with TCF1⁺CD8⁺ TR and other indicated pairs, with ns and exact *p*-values annotated above brackets from two-tailed Wilcoxon tests. **b** Mean distance to the nearest melanoma cell (μm) for indicated CD4⁺ and CD8⁺ subsets, grouped by treatment status (untreated, resistant, responder) with box-and-whisker plots and per-sample points; significant pairwise differences are labeled with *p*-values from two-tailed Wilcoxon tests. Box plots in **a**, **b** show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to 1.5 × the interquartile range. Individual points represent individual samples. **c** Schematic illustrating the shortest-distance relationships between melanoma cells and nearby CD4⁺ TR or CD8⁺ TR cells. Created in BioRender: https://BioRender.com/m3021b6.

ICI-responsive tumors, DC3 subsets exhibited exclusive interactions with cytotoxic TR populations, particularly GZM⁺ TR, NKG7⁺ TR, and CXCL13⁺ TR, as well as with CD4⁺ NKG7⁺ TR cells (Fig. 10e). These putative interactions were dominated by DC3-derived chemokine and cytokine signals linked to T cell recruitment and activation.

Specifically, DC3 cells expressed the highest levels of CCL17 and CCL19 in responders (Fig. 10f and Fig. S11d), while their corresponding receptors CCR4 and CCR7 were upregulated on TR subsets (Fig. 10g and Fig. S11c, e). CellPhoneDB modeling confirmed that CCL19–CCR7 and CCL17–CCR4 ligand–receptor pairs were significantly enriched between DC3 and TR cells in responders compared with resistant or untreated tumors (Fig. 10h, i and Fig. S11d, e), suggesting enhanced chemotactic guidance and localization of TR cells by DCs in the ICI-responsive microenvironment.

We further analyzed the IL15–IL15R signaling axis, which is critical for TR maintenance and activation[48]. Expression of IL15 was highest in DC3 cells of ICI-responsive tumors (Fig. 10j), with concurrent upregulation of IL15RA, FCGR2A, and IL2RB receptor subunits on TR cells (Fig. 10k and Fig. S11f, g). Correspondingly, IL15–IL15R interactions were strongly enriched between DC3 cells and both CD8⁺ and CD4⁺ TR subsets in responders (Fig. 10l), indicating that DC3 cells may act as key activators of TR function within the triad.

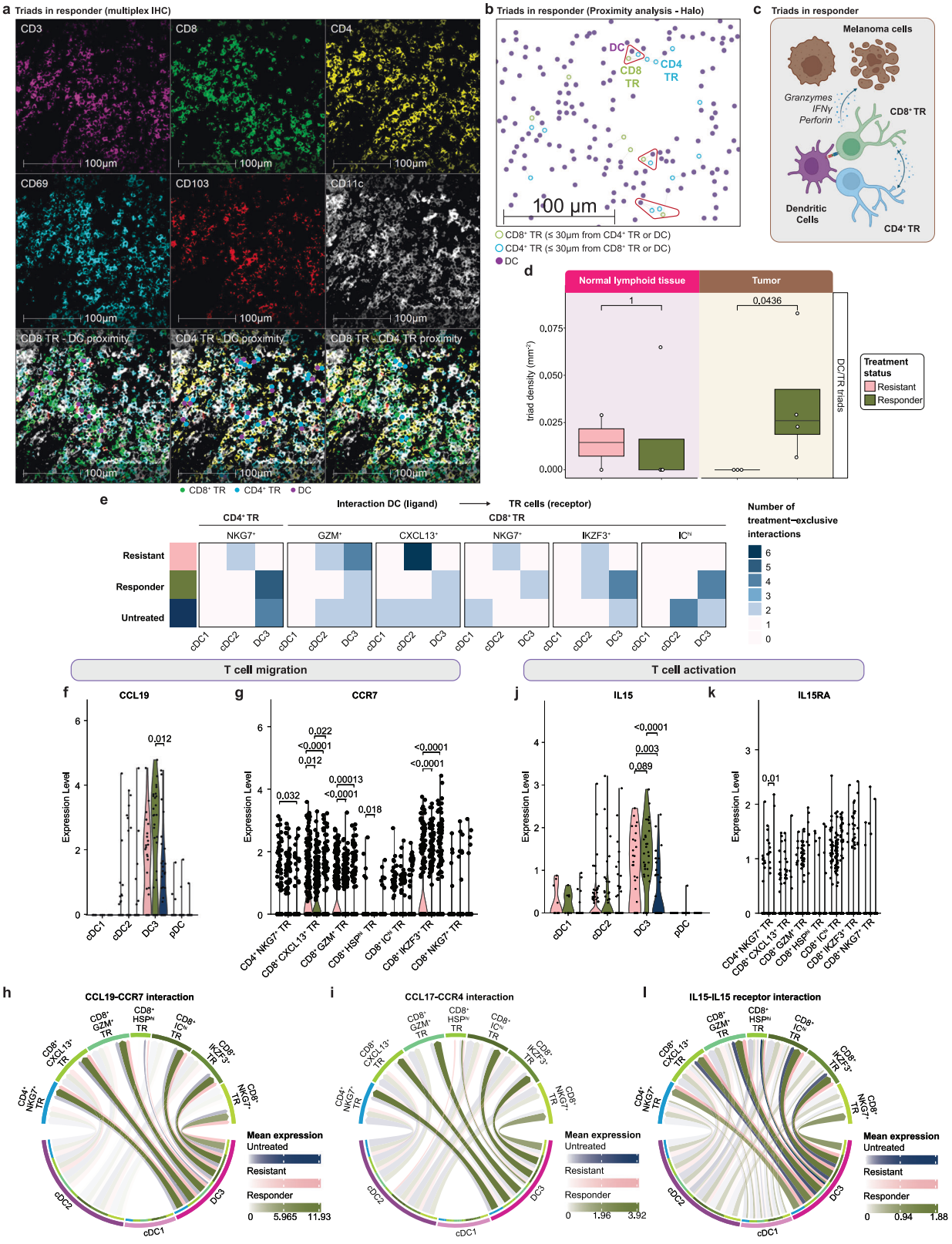
Together, these findings demonstrate that CD8⁺ TR–CD4⁺ TR–DC3 immune triads are a defining feature of ICI-responsive tumors. Within these structures, DC3 cells likely orchestrate TR localization via CCL17–CCR4 and CCL19–CCR7 chemokine gradients and may support TR effector function through IL15-mediated activation, consistent with coordinated, spatially integrated anti-tumor immunity.

Collectively, these data support a model in which coordinated spatial organization and molecular crosstalk between CD8⁺ TR, CD4⁺ TR, and DC3 cells are associated with effective immune checkpoint blockade therapy. This model defines an immune triad architecture consistent with integrated localization, activation, and effector function that may contribute to durable anti-tumor immunity.

Discussion

T cell differentiation remains an intensely investigated field in cancer, driven by the need to better understand which subsets are pivotal for ICI response. To date, several studies support the concept that progenitor-exhausted T cells (T_{Pex}) are the key population responding to tumor antigen presentation in draining lymph nodes and trafficking from the periphery into the TME[23]. In parallel, TR cells have also been associated with local immune surveillance, improved overall survival, and systemic response to ICI[10,13]. Whilst these findings may initially appear conflicting, an alternative interpretation is that they neglect distinct stages of T cell differentiation kinetics induced by ICI treatment within the TME. To interrogate this paradigm, we combined bulk- and single-cell RNA/TCR-Seq analyses with quantitative mIHC of pre- and post-ICI treated lymph node metastases, providing a high-dimensional, spatially resolved understanding of the role TR cells play in mediating ICI responses in melanoma. We showed that high levels of intratumoral CD8⁺ TR were associated with ICI responses. In particular, we observed that TCF1⁺ progenitor-like TR cells were enriched in ICI-responsive tumors, implicating this subset in sustaining long-term anti-tumor immunity. Furthermore, TR differentiation, persistence, and cytotoxic

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activation appeared to be coordinated by distinct TME cues, culminating in the formation of in situ CD8⁺TR-CD4⁺TR-DC triads, observed exclusively in ICI-responders. Our data also revealed CD4⁺ TR cells may also exert direct cytotoxic activity towards melanoma, and that increased IFN-γ response in melanoma cells during ICI-resistance may underlie why IFN-γ based biomarkers are imperfect predictors of treatment response.

In solid cancers, studies have shown that TCF1⁺ CD8⁺ T cells sustain anti-tumor immunity through continuous renewal within tumor-draining lymph nodes[49]. This reservoir of 'progenitor-like' T cells possesses self-renewal capacity and can differentiate into cytotoxic effector cells, replenishing those which may have become exhausted over time. The presence of TCF1⁺ CD8⁺ T cells has been associated with enhanced ICI response[23,50–52]; and in melanoma, TCF1⁺

Fig. 10 | Spatially organized CD8⁺ TR–CD4⁺ TR–DC immune triads and chemokine programs characterize ICI-responsive tumors. a mIHC from a responder tumor showing spatial triads composed of CD8⁺ TR, CD4⁺ TR, and DC (100 μm scale bars). Images are representative of independent samples from resistant ($n = 4$) and responder ($n = 4$) patients. **b** Proximity analysis (HALO) in a responder identifying triads where CD8⁺ TR are within ≤30 μm of CD4⁺ TR or DC, and CD4⁺ TR are within ≤30 μm of CD8⁺ TR or DC; symbols denote CD8⁺ TR (filled), CD4⁺ TR (open), and DC, with melanoma cells. **c** Schematic illustrating melanoma cells adjacent to CD8⁺ TR, CD4⁺ TR, and DC, highlighting cytotoxic programs (granzymes, perforin) and IFN-γ associated with these interactions. Created in BioRender: https://BioRender.com/nvac3ct. **d** Quantification of triad density per mm² across normal lymphoid tissue and tumor compartments, stratified by treatment status. Box plots show the median (center line), the 25th and 75th percentiles (box limits), and whiskers extending to 1.5 × the interquartile range. Individual points represent individual samples. Statistical significance was assessed by two-sided Wilcoxon signed-rank test. **e** Treatment-exclusive DC–TR interaction counts across treatment groups, with DCs as ligand-expressing cells and TR subsets as receptors; bars indicate ligand–receptor pairs unique to each group. **f** Mean expression of CCL19–CCR7 interaction across DC subsets (cDC1, cDC2, DC3) and TR subsets including CD4⁺ NKG7⁺ TR and CD8⁺ TR phenotypes (CXCL13⁺, GZM⁺, HSP^{hi}, IKZF3⁺, IC^{hi}). **g** Mean expression of CCL17–CCR4 axis across the same DC and TR subsets, showing treatment-group differences consistent with enhanced DC–TR communication in responders. **h**, **i** Violin plots showing chemokine expression and relative receptors associated with T-cell migration, including **h** CCL19 and **i** CCR7, across dendritic cell subsets and CD4⁺/CD8⁺ TR populations (CXCL13⁺, IKZF3⁺, NKG7⁺). **j** Mean expression of IL15–IL15 receptor interaction across DC and TR subsets, highlighting elevated availability in responder tumors. Expression of T-cell activation–related cytokines and receptors, including **k** IL15 and **l** IL15RA, across DC and TR subsets. Statistical significance was determined using pairwise Wilcoxon tests with adjusted *p*-values indicated above comparisons.

CD8⁺ TR cells displayed superior proliferative and IFN-γ-producing capacity[39]. However, a definitive clonal relationship between the progenitor and effector subsets in the context of ICI response has not yet been established. By integrating spatially resolved single-cell RNA/TCR-Seq data, our study provides evidence that CD8⁺-TCF1⁺ TR cells undergo clonal expansion and likely serve as a progenitor pool that differentiates into cytotoxic effector TR cells targeting melanoma cells.

While CD8⁺ TR cells are critical for ICI response in melanoma, we also observed marked enrichment of CD4⁺ TR cells in ICI-responders. CD4⁺ T helper cells can enhance the survival and differentiation of CD8⁺ TR cells during infections[53]. In our study, CD4⁺ TR cells shared approximately 25% of genes with T_{fh} cells, and closely interacted with CD8⁺ TR via the CCL4-CCR5 axis, which is crucial for mediating T cell migration, retention, and functional responses in tissues[54]. Additionally, in ICI-responders, this CD4⁺ TR population exhibited cytotoxic features, including upregulation of GZMB, GZMK, NKG7, and enhanced degranulation, though without evidence of extensive clonal expansion. Taken together, these findings suggest that CD4⁺ TR cells contribute to ICI response by supporting CD8⁺ TR function. However, the nature of their antigenic specificity, potential for direct cytotoxicity towards melanoma cells, and suitability as therapeutic targets require further investigation.

Beyond T cells, our study showed that multiple immune cell types including B cells, macrophages, and DCs cooperate to orchestrate the ICI response in melanoma. Specifically, DC3 cells interacted with both CD4⁺ and CD8⁺ TR cells to form an in situ immune triad exclusively in the TME of responder patients. DC3 cells provide key differentiation and survival signals to TR cells[26,27]. Consistent with this, our data support IL-15–mediated DC3–TR interactions, and IL-15 is known to be essential for the survival and maintenance of T_{RM} cells, particularly in the skin[46]. Under physiological conditions, IL15 signaling sustains the T cell memory pool through trans-presentation of DC-derived IL-15/IL-15Rα to IL-15Rβ/γc receptors on T cells, activating downstream JAK1/STAT3 and JAK3/STAT5 pathways[55]. To date, at least seven studies have reported spatial biomarkers of response to ICI in melanoma (e.g., density of PDL-1 + T cells or distance between PDL-1+ macrophages and cancer cells), but none have proposed validated thresholds which dichotomized patient outcomes[56]. The binary presence or absence of the CD8⁺ TR-CD4⁺ TR-DC immune triad, which clearly segregates responders from non-responders, represents a promising opportunity for a robust spatial biomarker predictive of response. Nonetheless, prospective clinical validation will be essential to confirm its predictive value.

Several biomarker signatures in melanoma have incorporated elevated IFN-γ signaling as a prognostic indicator[57] and/or predictor of ICI response[32,58]. However, our observation of high IFN-γ response in both immunosuppressive macrophages and melanoma cells in ICI-resistant tumors challenges the assumption that IFN-γ signaling universally promotes a favorable TME. In our study, ICI-resistant tumors were infiltrated by M2-like macrophages with an increased IFN-γ response gene signature, this was associated with an increased M2/M1 ratio indicative of a pro-tumorigenic and immunosuppressive TME milieu. Although IFN-γ typically promotes M1 polarization[59–61], sustained or excessive IFN-γ signaling can paradoxically lead to immunosuppression, T cell exhaustion and a tolerogenic TME[34,62,63]. Given the prolonged median interval between ICI treatment initiation and lymph node sampling in our cohort (Fig. S7a), it is possible that the observed sustained IFN-γ exposure may lead to feedback mechanisms that favor M2 polarization and contribute to T cell exhaustion, thereby facilitating tumor progression. Furthermore, melanoma cells in ICI-resistant tumors displayed significantly increased IFN-α and -γ signaling via IL-6/JAK/STAT3 and TNF-α.

These findings suggest that, under chronic inflammatory conditions, IFN-driven signaling likely contributes to an immunosuppressive environment that promotes immune evasion and therapeutic resistance.

There are limitations to our study. First, tumor samples were collected opportunistically during routine clinical care, introducing inherent variability in patient characteristics and treatment histories. The study is underpowered to determine systematic biases according to age, sex, and treatment types. However, the orthogonal and external validation of our findings demonstrated consistent immune-TME features across patient groups. Second, our primary analyses were correlative in nature. However, the degranulation assays showing autologous antigen recognition of tumor cells by TR cells provided further evidence for these cells as direct mediators of anti-tumor response under ICI. The archetype analysis focused on immune populations, reflecting our aim to define immune-mediated mechanisms of ICI response centered on tumor-resident T cells. The 8-marker mIHC panel was optimized for TR profiling, which limited accurate segmentation of non-immune compartments. Future high-plex spatial platforms could further resolve the spatial architecture of ICI response.

In conclusion, our findings highlight the critical role of diverse and enriched CD8⁺ TR cell interactions, particularly with CD4⁺ TR cells, DCs, and melanoma cells, in shaping a robust and effective anti-tumor immune response during ICI treatment. These findings underscore the complex interplay between multiple immune cell types within the TME and highlight the importance of non-lymphoid cells in shaping the immune landscape and therapeutic response in melanoma. Understanding these interactions could provide new avenues for developing biomarkers for response assessment as well as therapeutic interventions aimed at modulating the tumor microenvironment to enhance treatment efficacy.

Methods

Ethics

This retrospective study was approved by the Peter MacCallum Human Research Ethics Committee (13/141) and MRV (approved project MRV69).

Patient cohort

Lymph node tissue samples ($n = 32$) were provided by Melanoma Research Victoria (MRV) from 29 metastatic melanoma patients with stage III/IV melanoma who were treated at Peter MacCallum Cancer Centre between 2013 and 2023. 'ICI-resistant' tumors were collected from patients where surgery was performed in the context of progressive loco-regional lymph node metastases with or without prior ICI response in other organ sites. 'ICI-responsive' tumors were collected from patients in the context of either persistent FDG (fluorodeoxyglucose) uptake detected by positron emission tomography with evidence of a major treatment effect on histology, or where lymph nodes persisted in patients achieving a partial response by RECIST (Response Evaluation Criteria in Solid Tumors) (Supplementary Dataset 1). We received representative fragments of fresh, sterile tissue after surgery, from which we either froze tumor pieces or enzymatically digested the remaining tissues to obtain a single-cell suspension, which was then cryopreserved. Additionally, we accessed formalin-fixed, paraffin-embedded (FFPE) tissues through pathology.

Fourteen treatment-naïve samples were identified and classified as untreated. Patients treated with ICI ($n = 16$) received either monotherapy with anti-PD-1 (pembrolizumab or nivolumab) or combination therapy with anti-CTLA-4 (ipilimumab). Twelve patients displayed clinical features of ICI resistance and were classified as ICI-resistant. Six patients achieved a pathological response (tumor size reduction confirmed by imaging) and were classified as ICI-responders. A comprehensive summary of the clinical features, treatments and surgeries of this cohort are indicated in the Supplementary Dataset 1.

Bulk RNA-seq

Prior to RNA extraction, snap-frozen tissue samples were thawed in 10 volumes of RNAlater-ICE (Thermo Fisher Scientific, Massachusetts, USA) overnight at −20 °C. RNA was extracted using the RNeasy Mini Kit (Qiagen, Hilden, Germany) per the manufacturer's protocol. Each tissue piece was added to 350 µL of lysis buffer and manually homogenized using a mortar and pestle-like system, followed by QIAshredder (Qiagen) processing. The quantity and integrity of RNA samples were assessed using a TapeStation (Agilent, CA, USA) with RNA ScreenTape (Agilent) to determine the RNA Integrity Number (RIN). Paired-end poly(A) sequencing of the RNA samples was performed by the Molecular Genomics Core Facility at Peter MacCallum Cancer Centre on the Illumina NextSeq 2000 (Illumina, California, USA). RNA-seq libraries were generated using the NEBNext Ultra II Directional RNA Library Prep Kit for Illumina and the NEBNext Poly(A) mRNA Magnetic Isolation Module.

Bulk RNA-seq data were TMM-normalized and run through the removeBatchEffect algorithm via *limma* (Supplementary Methods 1a–d). Gene set z-scores were then calculated using *GSVA* for several curated gene sets. Differential expression analysis was performed between responder and resistant samples using a linear model via the *edgeR* and *limma* workflow, and log fold changes between these groups were used as ranks for a gene set enrichment analysis via *fgsea* using various sets from the Molecular Signatures Database[64].

Nanostring

RNA was isolated and purified from 10 µm tissue sections using the RNeasy FFPE Kit (Qiagen) according to the manufacturer's instructions. RNA was eluted from the columns in 20 µl of DEPC-treated water. RNA from validation samples was assessed for quality using a High Sensitivity RNA TapeStation assay (Agilent), applying a DV200 cutoff of >50% for downstream analysis. A total of 150 ng of RNA from each patient biopsy was analyzed using the Nanostring platform with the nCounter PanCancer Immune Profiling Panel (Nanostring; NS_CANCERIMMUNE_V1.1) and a custom TRM panel (Nanostring; PLS_TRM Panel).

Subsequent normalization and gene quantification were performed using the RUVIII algorithm from *ruv* (Supplementary Methods 2a–c)[65]. Samples were scored for TIS using a z-score gene set variation analysis from the *GSVA* package[66]. Differential expression analysis was performed between responder and resistant samples using a linear model via the *edgeR* and *limma* workflow[67].

Single-cell RNA-seq and TCR-seq

Cryopreserved cells were prepared for single-cell sequencing as previously described[68]. To enhance sample viability, the Dead Cell Removal Kit (Miltenyi Biotec) was employed following the manufacturer's protocol. Briefly, cells were incubated with Dead Cell Removal MicroBeads in Binding Buffer for 15 minutes at room temperature. The cell suspension was then applied to an MS MACS® Column placed in a magnetic separator. Viable cells were collected by allowing them to pass through the column, while dead cells remained bound. The enriched viable cell fraction was washed with Binding Buffer and counted using a hemocytometer.

For scRNA-seq and scTCR-seq, cell concentration was adjusted to 2,000 cells/µL in PBS with 0.04% BSA. Single-cell capture and library preparation were performed using the 10x Genomics Chromium 5' V2 Cell Capture Kit (single-cell RNA-Seq) or 10X Genomics 5' VDJ Enrichment kit (single-cell TCR-Seq) according to the manufacturer's instructions. This approach enables simultaneous profiling of gene expression and TCR sequences from the same cells.

Sequencing libraries were quantified using High Sensitivity RNA TapeStation assay (Agilent) and sequenced on an Illumina NovaSeq S4 platform to achieve a minimum depth of 25,000 reads per cell for gene expression and 5,000 reads per cell for TCR libraries. Data processing, including alignment, gene expression quantification, and TCR clonotype assignment, was conducted using the Cell Ranger pipeline. Rigorous quality control confirmed high data quality across samples, with consistent mitochondrial read content, RNA counts, and hybrid scores after filtering (Supplementary Methods 3a–c).

CellRanger outputs were analyzed downstream using *Seurat*[69,70]. Samples were first processed individually, where they were filtered to cells with more than 200 RNA features, and doublet-scored via the *scds* package[71]. *Azimuth*[72] projection and *SingleR*[73] annotation was also completed at the per-sample level, to assist with downstream annotation.

All samples were combined for downstream analysis and annotation. Samples were further filtered at this stage to remove cells with higher than 5% mitochondrial gene content, and to remove cells with a *scds* hybrid score greater than 1. To mitigate sample-driven clustering effects, we integrated the data using the *harmony* approach[74]. Seurat was used to produce a UMAP for this combined dataset and for identification of clusters via the *SLM* algorithm.

While initial cell-type demarcations were clear at a low resolution for the entire dataset, we split the data into broad subsets for higher resolution clustering and cell-type annotation. Subsets included DCs, B lymphocytes, macrophages, melanoma cells, as well as T lymphocytes and NK cells. In some cases, such as for T lymphocytes, further subsetting occurred for finer (i.e. CD4+ and CD8 + T cell-specific) annotations. For consistency, clustering resolution was set to 1 in all subset analyses. Cell-type annotation was informed primarily by cluster markers as determined through Seurat, as well as being guided by preliminary annotations at the per-sample level, and *SingleR*- and *Azimuth*-predicted annotations. Lastly, we derived cell-wise module scores exhaustion and progenitor of exhaustion gene sets[40], using Seurat's AddModuleScore function.

The number of cells assigned to each finalized cell-type annotation on a per-sample basis was used to conduct differential abundance testing via the *edgeR* package's glmQLFit model, to determine enriched cell-types across treatment status. We also used cell-type annotations in comparison of mean module scores for certain gene sets across treatment status. The TCR sequencing data from our single-cell analysis was also used at this downstream stage, where we characterized the cell-types composing expanded clones. Clones were defined as cells from the same sample sharing an identical amino acid CDR3 sequence for both the TCRA and TCRB genes. Clone sizes of 2, 10, and 100 provided the cutoffs for clones described as small, expanded, or hyperexpanded, respectively.

Flow cytometry

Fresh lymph node tissue samples were digested to obtain a single-cell suspension. For tumor digestion, tumor fragments were suspended in a digestion mix (RPMI, Collagenase IV, DNase I) for 30–60 minutes. After removing the supernatant, cells were lysed with red blood cell (RBC) lysis buffer (BD BioSciences, NJ, USA) before being resuspended in FACS buffer (2% BSA in 1XPBS 1X).

Cells were washed and stained with LIVE/DEAD Near-IR viability dye (Invitrogen, CA, USA) at a final dilution of 1:4,000 in phosphate-buffered saline (PBS) for 5 minutes at RT. Surface markers were then stained with an antibody cocktail (Supplementary Table 4) and incubated in the dark at 4 °C for 30 minutes. Cells were then centrifuged to remove the supernatant and fixed with 2% paraformaldehyde (PFA) for 10 minutes at RT.

For intracellular staining, cells were fixed using eBioscience Fix/Perm (Invitrogen) and washed with eBioscience PermWash (Invitrogen) at a 1:10 dilution in water, according to the manufacturer's specifications. Cells were then centrifuged and stained with intracellular markers (Supplementary Table 4) for 30 minutes at 4 °C. Fixation buffer was removed by centrifugation, and cells were resuspended in FACS buffer prior to acquisition.

UltraComp eBeads Compensation Beads (Invitrogen) were used to set up single-color compensation controls. An isotype control panel was separately stained for each sample. All data were acquired on a BD FACSymphony A5 (BD Science) and analyzed using FlowJo software (v10.8.1, BD Science).

Degranulation assay

To assess antigen-specific cytotoxic activity of tumor-infiltrating lymphocytes (TILs), we performed a short-term co-culture degranulation assay using autologous melanoma cells as targets. Cryopreserved TILs and melanoma cells from untreated, ICI-resistant and ICI-responder patients were thawed and washed in complete RPMI 1640 medium supplemented with 10% FBS and 1% penicillin/streptomycin. Autologous melanoma cells were plated at 1×10^5 cells per well in 48-well plates and incubated overnight. The following day, TILs were resuspended in cytokine-free complete RPMI at 2×10^5 cells per well and co-cultured with melanoma cells at a 2:1 effector-to-target (E:T) ratio for 4 hours at 37 °C.

Four conditions were tested per donor: (1) TILs alone, (2) TILs + PMA/ionomycin (positive control; final concentrations 1 ng/mL PMA and 1 μg/mL ionomycin), (3) TILs + melanoma, and (4) TILs + melanoma + IL-2 (6,000 U/mL). All conditions included GolgiStop (BD Biosciences; 1×) to prevent cytokine secretion. During co-culture, CD107a-APC-H7 antibody (BD Biosciences, 1:50 dilution) was added directly to each well to monitor surface mobilization of lysosomal-associated membrane protein-1 (LAMP-1) as a marker of degranulation.

After incubation, cells were harvested, washed in PBS, and stained with a viability dye followed by surface staining for CD3, CD4, CD8, CD69, CD103, and CD39. For intracellular detection of Granzyme B, cells were fixed and permeabilized using the eBioscience Fixation/Permeabilization kit according to the manufacturer's protocol and stained with anti-GZMB antibody. Samples were acquired on a Cytek Aurora flow cytometer, and data were analyzed using FlowJo v10.8.1 (BD Biosciences).

Degranulation was quantified as the percentage of $CD107a^+$ cells within gated $CD8^+$ or $CD4^+$ $CD39^+CD69^+CD103^+$ populations. Statistical analysis was performed using unpaired Mann-Whitney U-test.

OPAL mIHC

All melanoma lymph node metastasis samples were formalin-fixed and paraffin-embedded (FFPE). Subsequently, serial 3-μm sections were cut, and H&E-stained slides from each FFPE block were evaluated by a pathologist, who defined "tumor" and "normal" tissue regions. The adjacent 4-μm FFPE tissue section from each biopsy was mounted on a Trajan Series 3 adhesive slide. Slides were then deparaffinized, rehydrated, underwent antigen retrieval, and were stained by serial passage, as previously described[75].

Slides were stained using either a TR-plus-melanoma panel, including DAPI, CD3, CD4, CD8, CD69, CD103, TCF1, and SOX10, or a TR-DC interaction, including DAPI, CD3, CD4, CD8, CD69, CD103, CD11c, and SOX10 (Supplementary Table 3). All slides were subsequently imaged using the Vectra Polaris™ Automated Quantitative Pathology Imaging System (Akoya Biosciences, Inc.). Slides were scanned at 20× magnification (resolution: 0.5 μm per pixel) using Phenochart v1.1 (Akoya Biosciences, Inc.).

Images were spectrally unmixed using inForm Tissue Analysis Software v2.6.0 (Akoya Biosciences, Inc.). Tissue and cell segmentation were performed using the Highplex v4 module in the HALO® software (Indica LABS – v3.5). An overview of the multiplex IHC single stain is provided in Fig. S9a.

Automated thresholding of mIHC data

The AMMIT model[76] was used to convert marker intensities of DAPI-positive cells as determined by the HALO software into cell phenotypes. AMMIT uses a mixture of skew-normal distributions model to call probabilistic thresholds trained on a subset of manually-thresholded samples. In this case, samples were split into tumor and residual lymph node regions where appropriate, both to decrease computational load and to control for autofluorescence differences between regions. The training and validation of the AMMIT model was performed on manually-thresholded samples, and the unthresholded samples were evaluated using the trained parameters from these steps. Cell phenotypes as indicated by the AMMIT model were used in downstream spatial analyses.

Spatial analysis

Individual sample spatial analysis was first run using the *SPIAT* R package as a framework[18]. We used *SPIAT* for the calculation of minimum distances between cell types, as well as for the identification of immune cell neighborhoods by *DBScan* clustering[77]. These results, as well as cell phenotypes, were then collated for a cohort-level analysis. Cellular densities were derived using areas calculated by the HALO software both for total tissue and tumor or non-tumor regions. Minimum distance data was averaged for each sample to provide mean distances between specific cell types. Minimum distance data from each cell, where under 30μm, identified close interactions for comparison of interaction frequency. This paradigm was also used to identify triads in the second OPAL panel, whereby triads were defined as triplets of cells each interacting with the others at less than 30μm.

Immune cell neighborhood data were also collated. The cellular composition of each neighborhood was used to inform a 100-nearest neighbors network using the *uwot* R package[78]. Leiden clustering of the adjacency graph for this network was used to identify clusters of similar neighborhoods, and permutation testing was performed on the cluster assignments to confirm that their cellular compositions significantly deviated from a null distribution. Clusters were then

categorized into archetypes based on the cluster's mean cellular composition and cell-type enrichment, where a composition >40% was denoted as 'Dominated', and >20% as 'Mixed'. A UMAP reduction of the neighborhood compositions was also produced for visualization of the archetypes. The *edgeR* package[79,80] was then used to conduct differential abundance tests, via the *glmQLFit* function, treating the number of each neighborhood archetype per sample as count data with respect to treatment status.

Cell interactome analysis

The *CellPhoneDB* algorithm[17] was used to analyze the interactomes of each of the scRNA-Seq samples. Each sample was restricted to cells with cell-type annotations of interest where more than 5 cells of that type were present, before being analyzed by *CellPhoneDB* for significantly enriched interactions. Each sample's results were then collated for cohort-level comparisons of interactions between various cell types. Specific interactions were compared using the complete (i.e. inclusive of non-significant) means outputs. Interactions were identified as unique to one treatment status where they were exclusively significantly enriched in one or multiple samples from that group.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The processed sequencing data have been deposited in the Gene Expression Omnibus (GEO) database: series GSE294272 for bulk RNA-Seq, series GSE294441 for Nanostring, series GSE294273 for scRNA-Seq, and series GSE294274 for TCR-Seq. Given the human origin of the samples, the raw data are available on the European Genome-Phenome Archive (EGA; accession number EGAS50000001722), and will be released by the lead contact upon request (email: paul.neeson@petermac.org). Raw primary imaging data and any additional information required to reanalyze the data reported in this paper are available from the lead contact upon request. Source data are provided with this paper.

Code availability

AMMIT code was developed by P.C. and deposited in the GitHub repository: https://github.com/neeson-lab/AMMIT.

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Author contributions

P.J.N. designed the study. A.D.P. and L.A. conducted experiments and performed the analysis. A.P. performed flow cytometry; A.P., T.N., S.M. and H.C. performed multiplex IHC, Nanostring and Bulk RNA-Seq experiments; A.A. performed the degranulation assays; A.D.P. performed scRNA-seq and scTCR-seq experiments; P.C., N.T. and R.Z. performed all bioinformatics analyses. T.M.H. performed tissue mark-ups on H&E slides. M.B., P.S., R.S., M.P.L., G.Y., L.S., A.D.R., S.S., D.E.G., G.A.M. and L.M. curated the data. S.M., K.B.C., and K.W. from MRV organized patient consent and tissue collection. L.A. and J.C. curated the clinical features of the cohort. A.D.P., L.A. and P.J.N. created the figures and wrote the manuscript with input from all authors. All authors have read and agreed to the published version of the manuscript. P.J.N. acts as guarantor for the study.

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Competing interests

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